

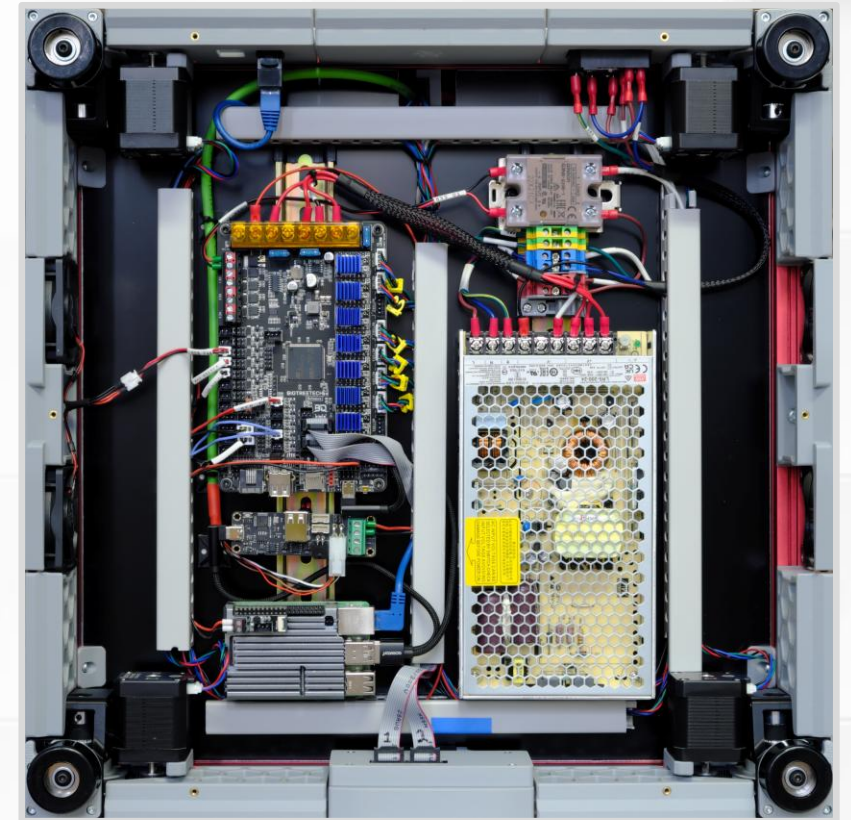
# *An Introduction to 3D Printing*

## *Part 1: FFF Printers*

*Process, Limitations, Dimensional Accuracy, and  
Advantages*

# About Me

- Hobby woodworking.
- Built my own printer to learn 3D printing.
- Voron 2.4 with over 6k hours of print time.



([Voron 2.4 GitHub](#))

# Websites

- 3D designs.  
[printables.com/@FlyingGyroscope](https://printables.com/@FlyingGyroscope)
- Get this guide.  
[github.com/FlyingGyroscope](https://github.com/FlyingGyroscope)



# Goals & Audience

- Training guide for students.
- Learn *strengths* and *weaknesses* of 3D printing.
- Make great prints.



# Outline

## **Part 1: Printers**

Process

Limitations

Dimensional Accuracy

Advantages

## **Part 2: Practical Application**

Examples

Critical Thinking

Design

## **Part 3: Science**

Vibrations and Waves

Linear Motion

Thermal Expansion

# 3D Printing is a Process

## 1) Start with a 3D design.

- Check the license for restrictions when you download.
- *STL*. Approximates shapes with triangles.
- *STEP*. Real geometry. Great for 3D modeling.
- *3MF*. Compressed container, like a zip file.  
Multiple objects, slicer settings, multi-color settings, ...

## 2) Slice and process the design with software.

- Printers do not accept 3D models!
- “Slice” objects into vertically-stacked layers.
- Create G-code file. List of toolpaths and commands that encode:  
Coordinates, speeds, temperatures, ...

# 3D Printing is a Process

## 3) **Print.**

- Do you need to change the design?
- Change print settings?

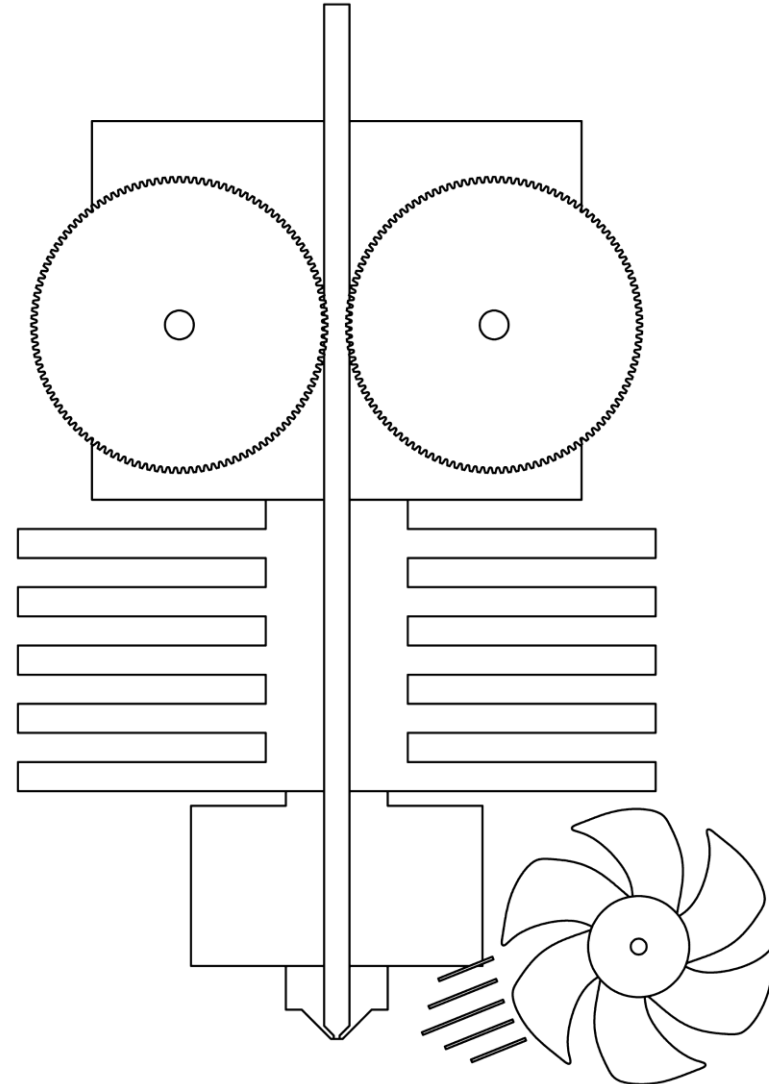
## 4) **Post-process.**

- Remove supports.
- Assemble.
- Glue.
- Sand or file.
- Paint.



# Pushing Plastic

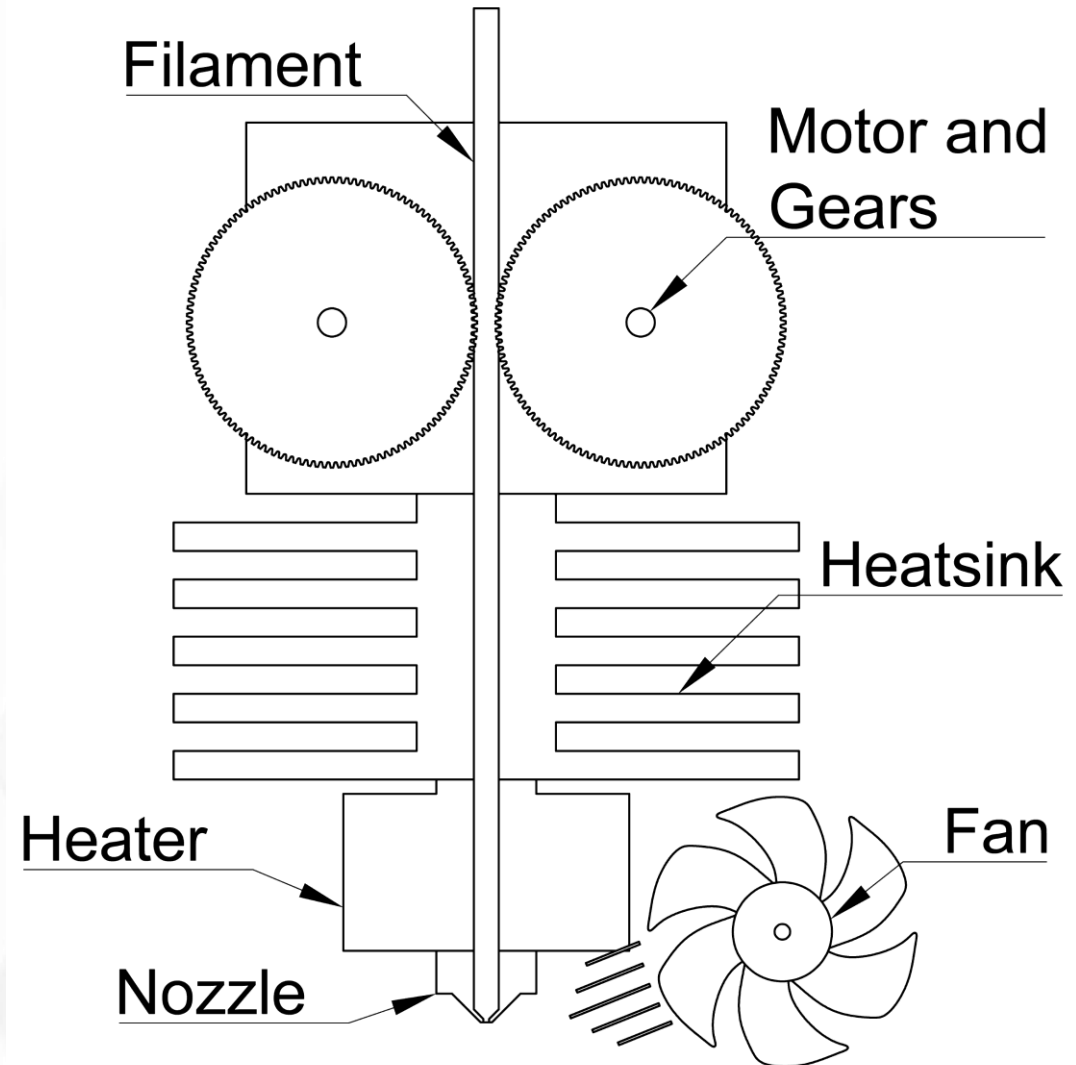
- What kind of 3D printing?
- Filament and extrusion-based additive manufacturing.
- **FFF. Fused filament fabrication.**
- FDM™. Fused deposition modeling.
  - Fundamentally the same but owned by Stratasys.
- Toolhead. Moves around and deposits plastic.





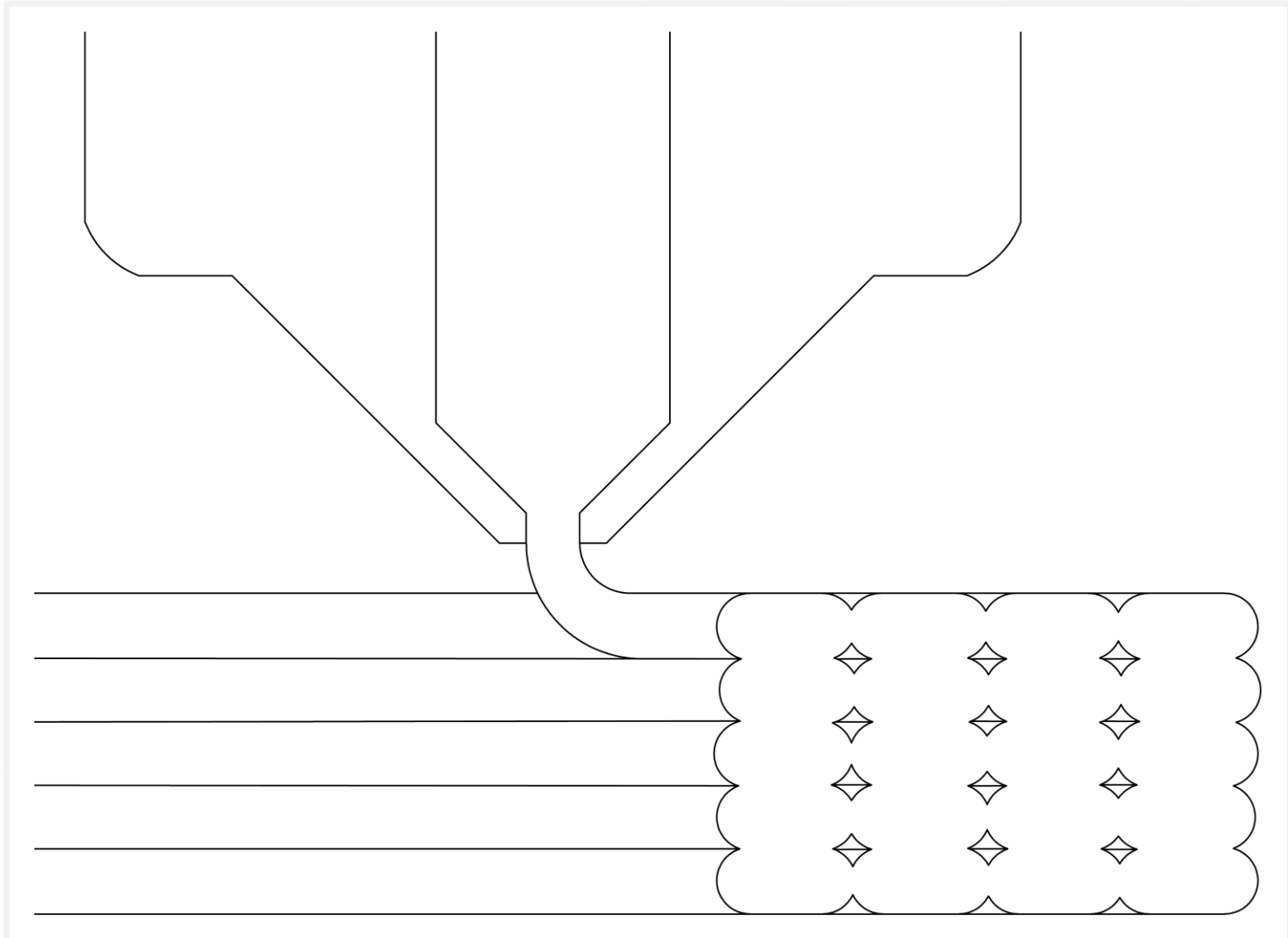
# Pushing Plastic

- Filament. Long, continuous plastic rod.
- Motor and gears grab and push filament.
- Heatsink keeps heat in the hotend (heater + nozzle).
- Heater softens filament at the glass transition temperature.
- Nozzle opening shapes plastic.
- Fan cools hot plastic.



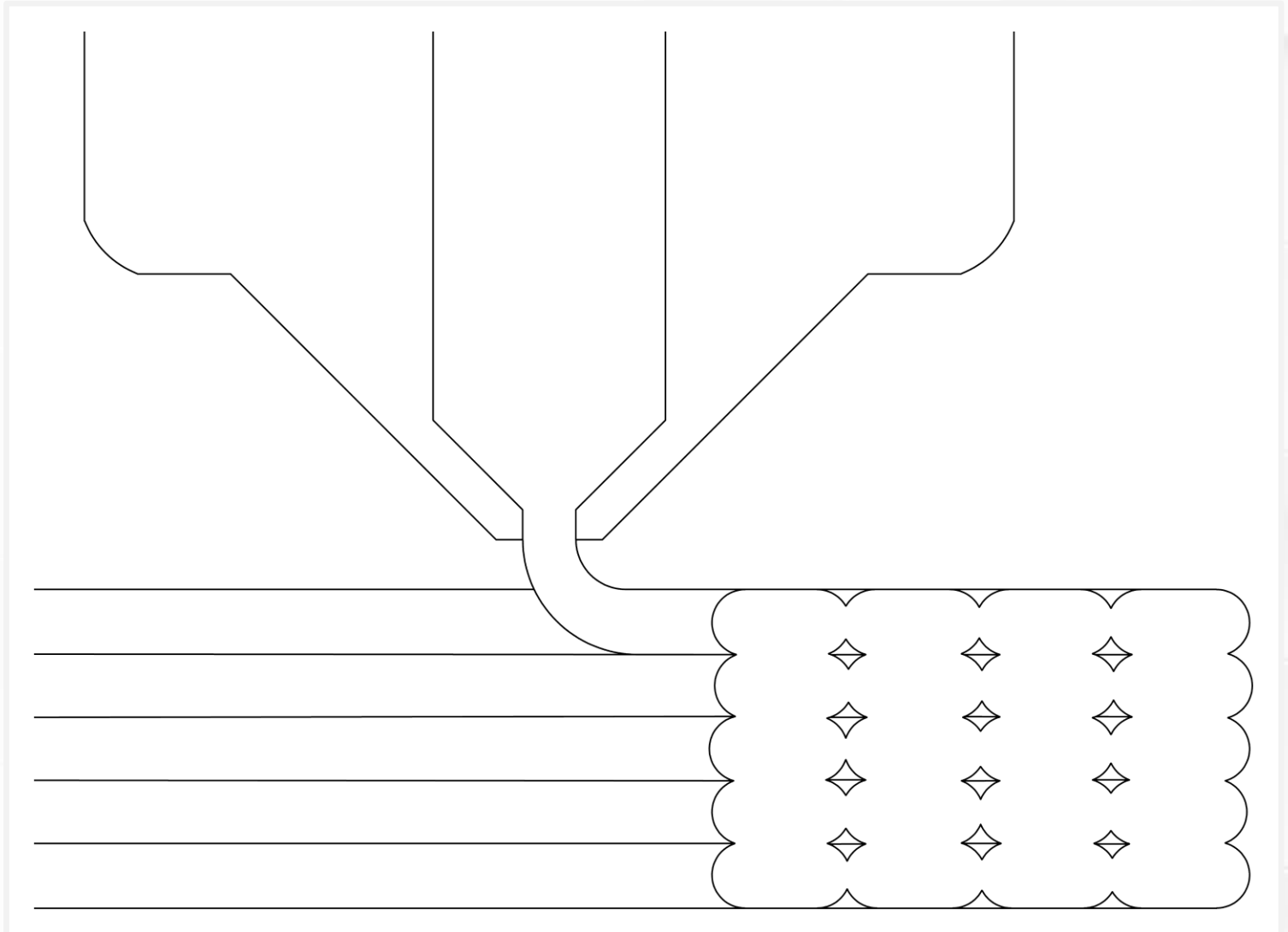
# Extrusions

- Toolhead deposits lines called **extrusions**.
- Best to squish extrusions into place.
- Lock in place as they cool.



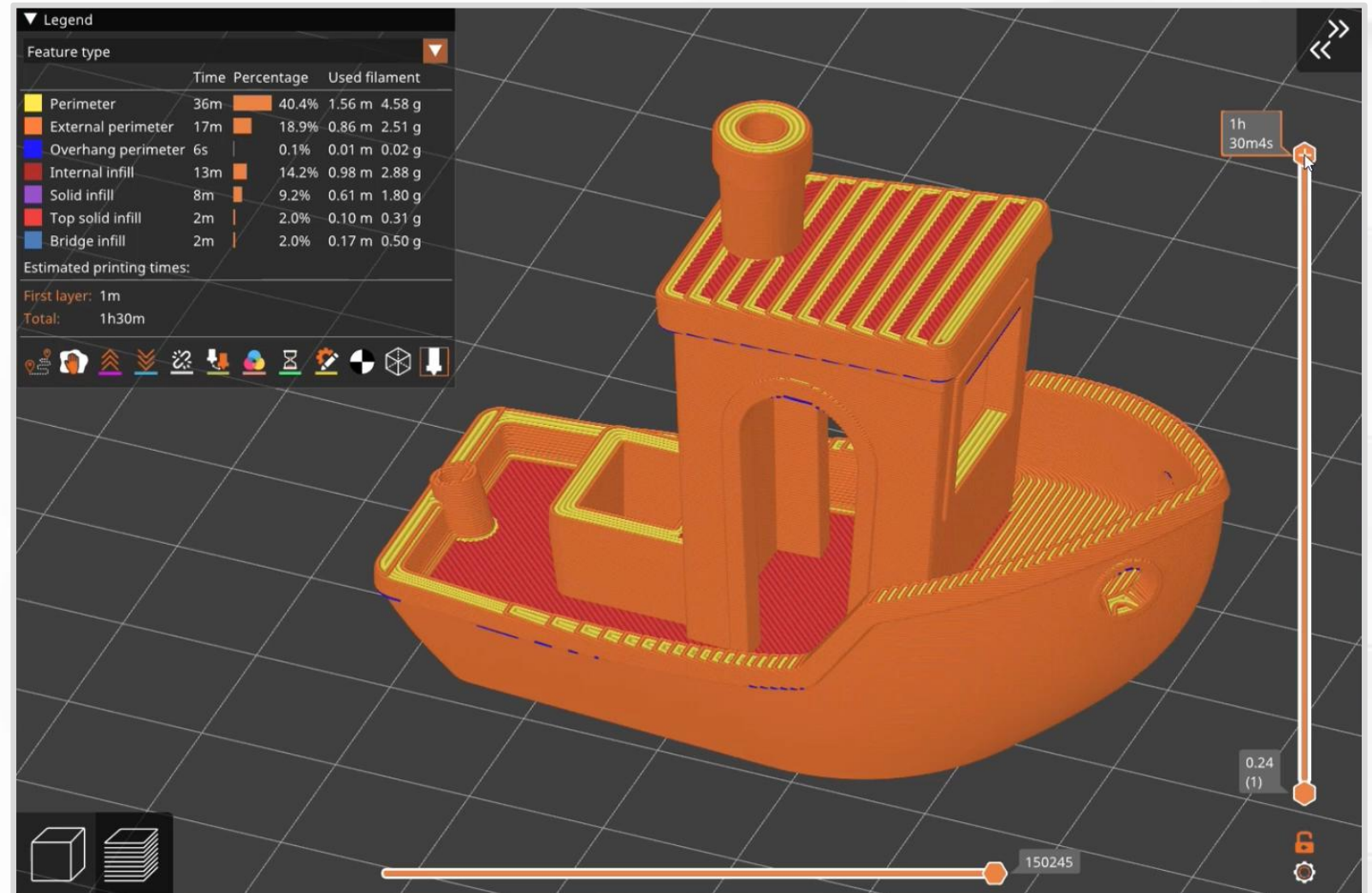
# Stacking Layers

- Printers *simulate* 3D shapes by stacking 2D layers.
- You could call it 2.5D printing.



# Slicer Layers

Let's see how PrusaSlicer handles layers.



(Animation,  
FlyingGyroscope GitHub)

# Slicer Layers

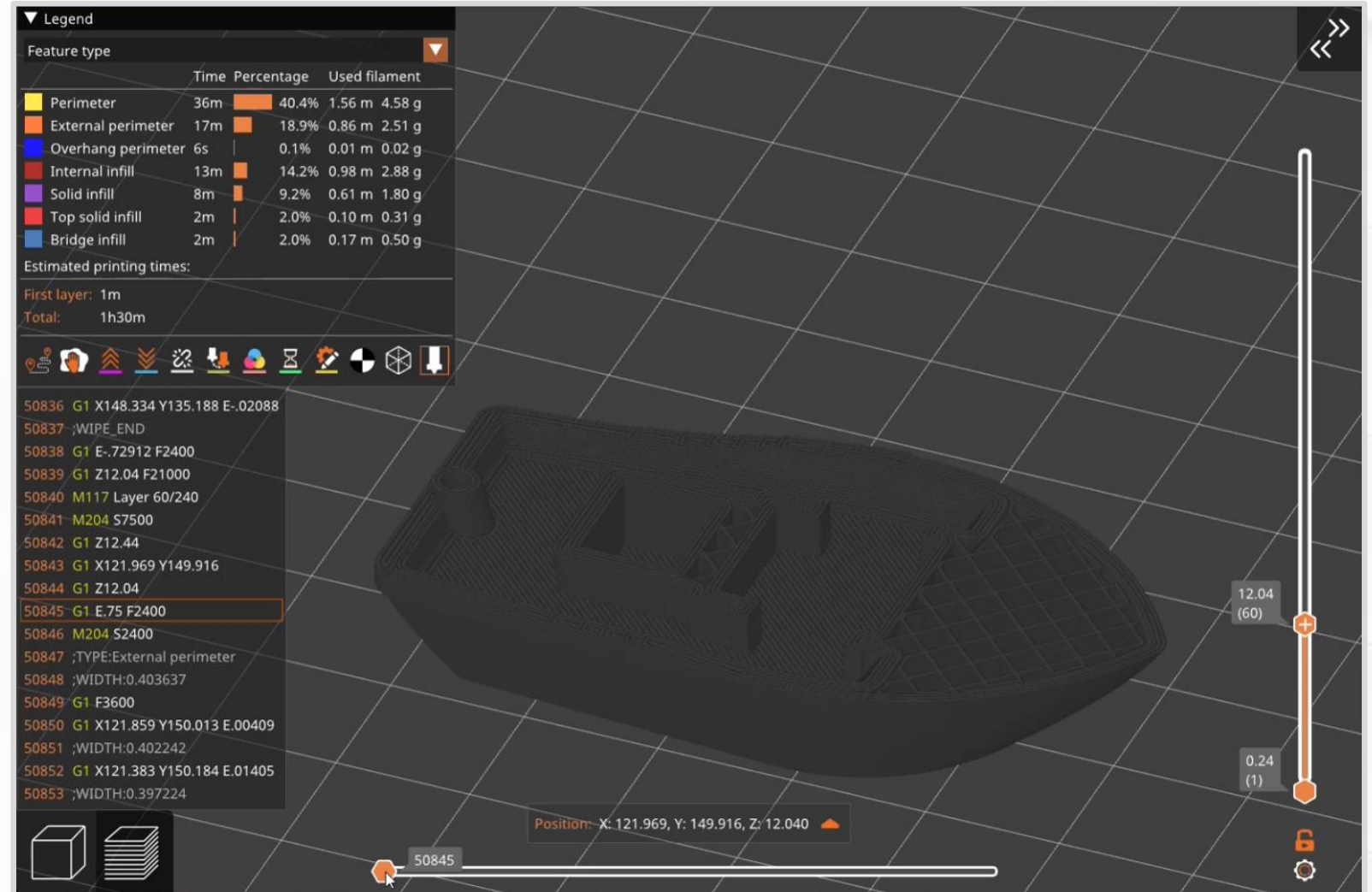
- Sort extrusions by shape and geometry.
- Color-coded.
- Adjust settings for each category.

| ▼ Legend                                                                             |                    |      |                                                                                            |               |        |
|--------------------------------------------------------------------------------------|--------------------|------|--------------------------------------------------------------------------------------------|---------------|--------|
| Feature type ▼                                                                       |                    |      |                                                                                            |               |        |
|                                                                                      |                    | Time | Percentage                                                                                 | Used filament |        |
|    | Perimeter          | 36m  |  40.4%  | 1.56 m        | 4.58 g |
|    | External perimeter | 17m  |  18.9%  | 0.86 m        | 2.51 g |
|    | Overhang perimeter | 6s   |  0.1%   | 0.01 m        | 0.02 g |
|    | Internal infill    | 13m  |  14.2%  | 0.98 m        | 2.88 g |
|    | Solid infill       | 8m   |  9.2%   | 0.61 m        | 1.80 g |
|   | Top solid infill   | 2m   |  2.0%  | 0.10 m        | 0.31 g |
|  | Bridge infill      | 2m   |  2.0% | 0.17 m        | 0.50 g |
| Estimated printing times:                                                            |                    |      |                                                                                            |               |        |
| First layer: 1m                                                                      |                    |      |                                                                                            |               |        |
| Total: 1h30m                                                                         |                    |      |                                                                                            |               |        |

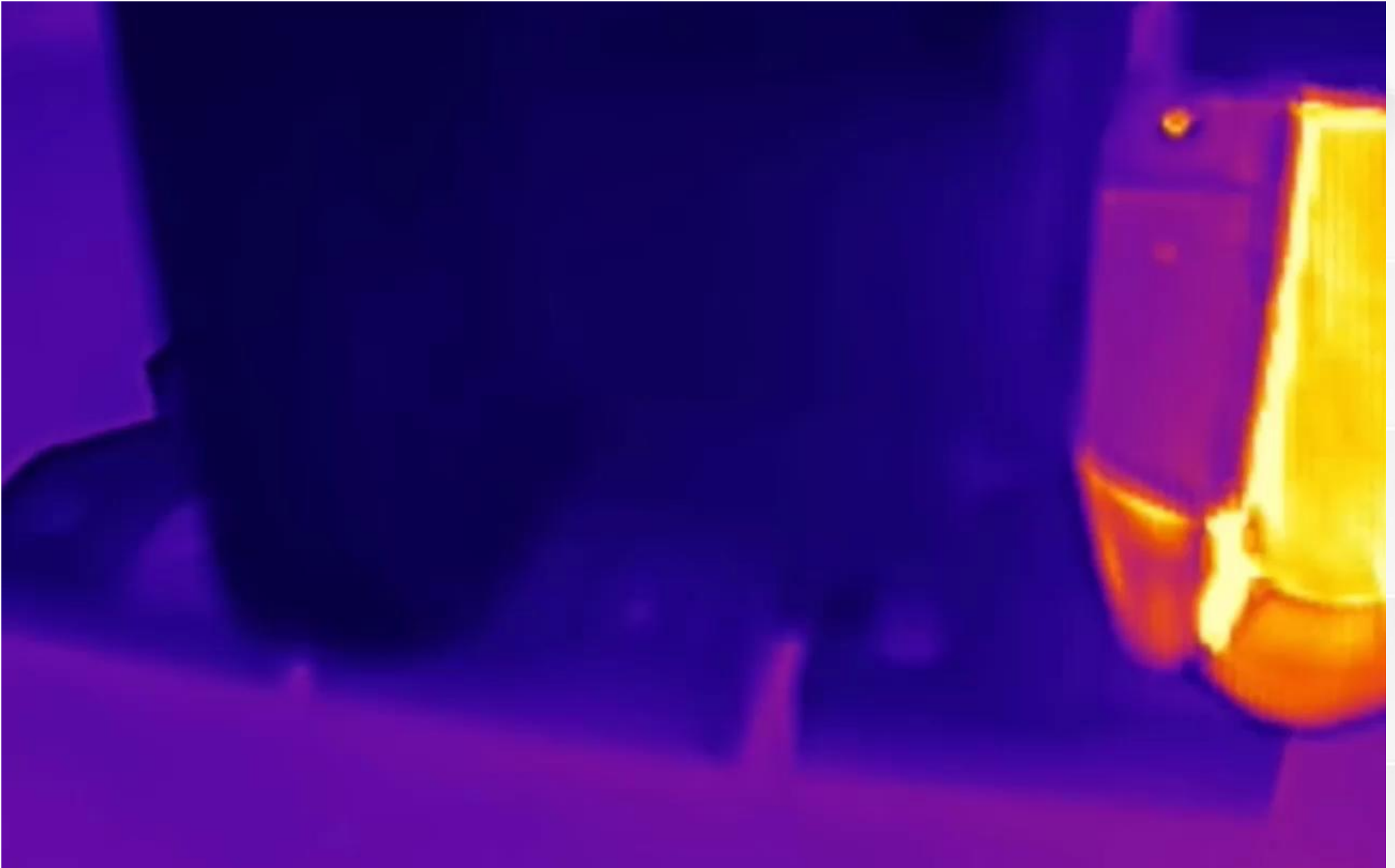
# Slicer Layers

- Single layer preview.
- G-code shows what the printer is actually doing (lower left).
- *G1* = Move. Include X, Y, or Z coordinates.
- *G1 X Y E* = Move and extrude.

(Animation,  
FlyingGyroscope GitHub)



(Animation,  
Thermal camera  
video by  
NeedItMakelt)





# Easy Filaments

- **PLA** (derived from corn starch).
  - General-purpose filament for indoor applications.
  - Excellent bridging.
  - Very stiff.
  - Creeps. Deforms permanently under load.
  - Low temperature resistance. Softens inside a car on a hot summer day.
- **PETG** (plastic water bottles).
  - Pretty good general-purpose filament.
  - Hygroscopic. Needs low-humidity storage.
  - Elastic. Excellent choice for springs.
  - Moderate temperature resistance. Can be used outdoors.

# Easy Filaments

- What makes PLA and PETG easy?
  - Low warp. Low shrink.
  - High dimensional accuracy.
  - Room temperature print chamber. Open-air, non-enclosed.

# Proper Protocol

Let's review some everyday guidelines.

- Secure the loose end of a spool of filament at all times.
  - Tangles ruin prints and waste time.
- Do not reach into the printer while it is running.
  - Use touchscreen or buttons to pause and resume prints.
- Watch the first layer before you walk away.
- Avoid burns.
  - Read nozzle and print bed temperatures on the display.
  - Never reach near a hot nozzle (210 °C = 410 °F).
- Do not leave fingerprints on print sheets.

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Vibrations and Waves

Linear Motion

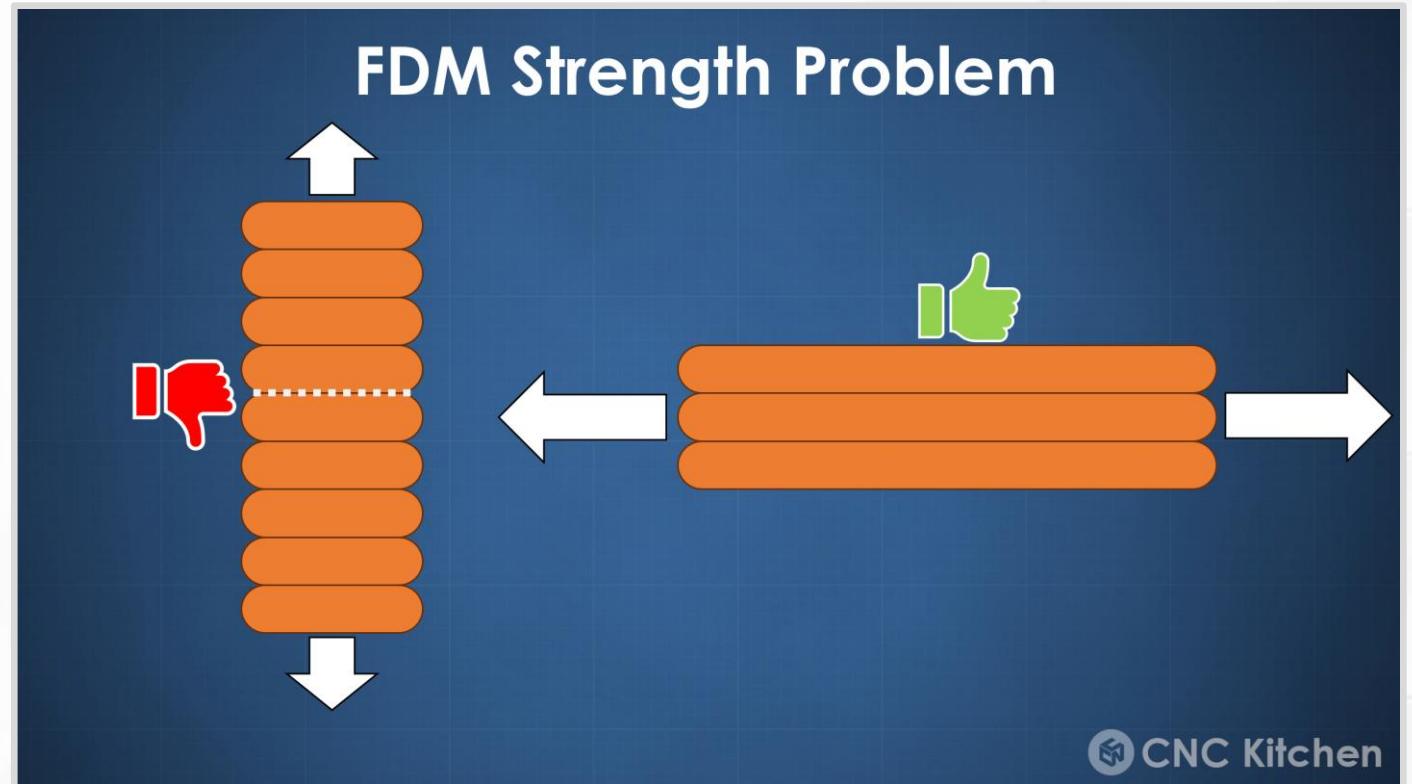
Thermal Expansion

# Thermal Cycle

- Some plastics shrink as they cool.
  - Up to 1 or 2%.
  - Seems small, but large enough to distort dimensions.
  - Leads to cracking and delamination in extreme cases.
- Advice?
  - Tightly control ambient temperature and cooling rate.
  - Some filaments (ABS, Nylon) like it hot → Enclosed print chamber, extra heating.
  - Some filaments (PLA) like it cold → Open-air printing, extra cooling.
  - PETG is somewhere in the middle.

# Anisotropic Strength

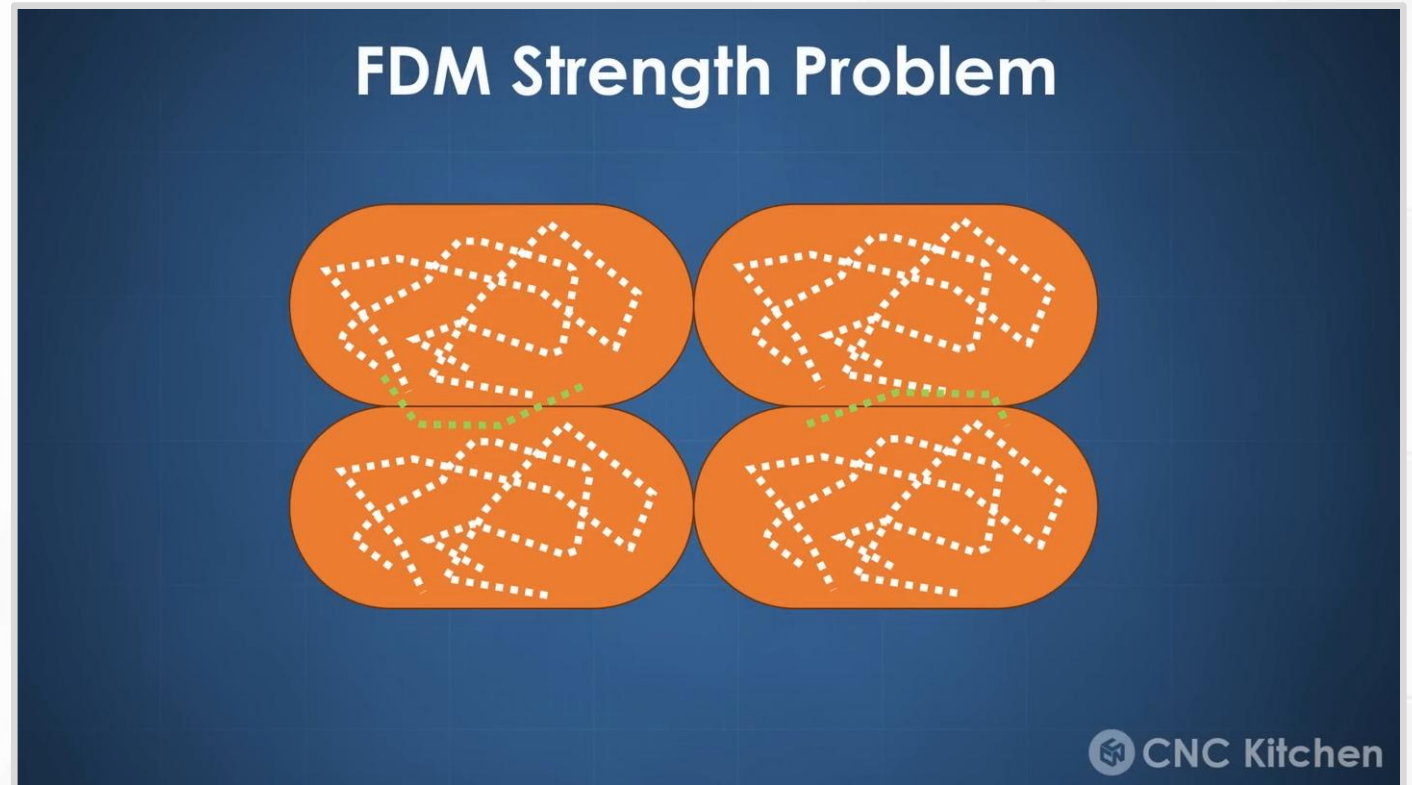
- Print orientation and layer lines affect strength.
- Layer-to-layer bonding is a weak point!
- And those weak points align.



([YouTube video](#) by CNC Kitchen)

# Anisotropic Strength

- Within an extrusion, hot filament bonds to hot filament.
- For a new layer, hot filament partially bonds to a cold layer.
- The “wrong” orientation can cost around 50% strength.



([YouTube video](#) by CNC Kitchen)



# Anisotropic Strength

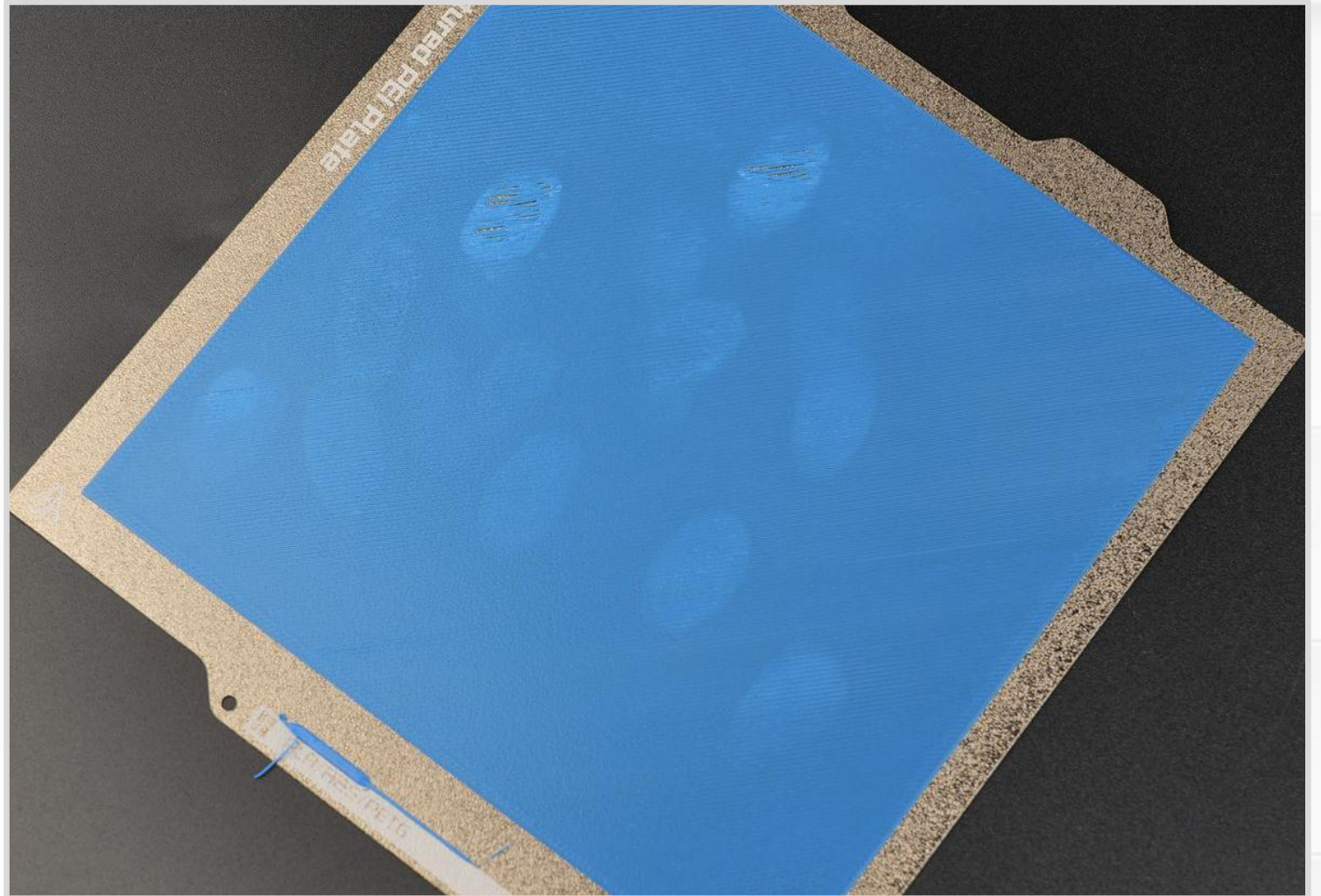
- Advice?
- Rotate prints to align layer lines and forces.
- Avoid tall and skinny shapes.
- Improve layer bonding.
  - Increase printing temps.
  - Decrease printing speed.

# First Layer

- What is the most important layer in a 3D print?
  - The first layer!
  - First layers keep everything locked in place.
- Advice?
  - Always watch the first layer.
  - Tune first layer squish (extremely sensitive).
  - Flat surface area on the bottom of the 3D object.
  - Heated print bed.
  - Perfectly clean print bed.
  - Apply adhesion promoters (glue).

# First Layer Mistakes

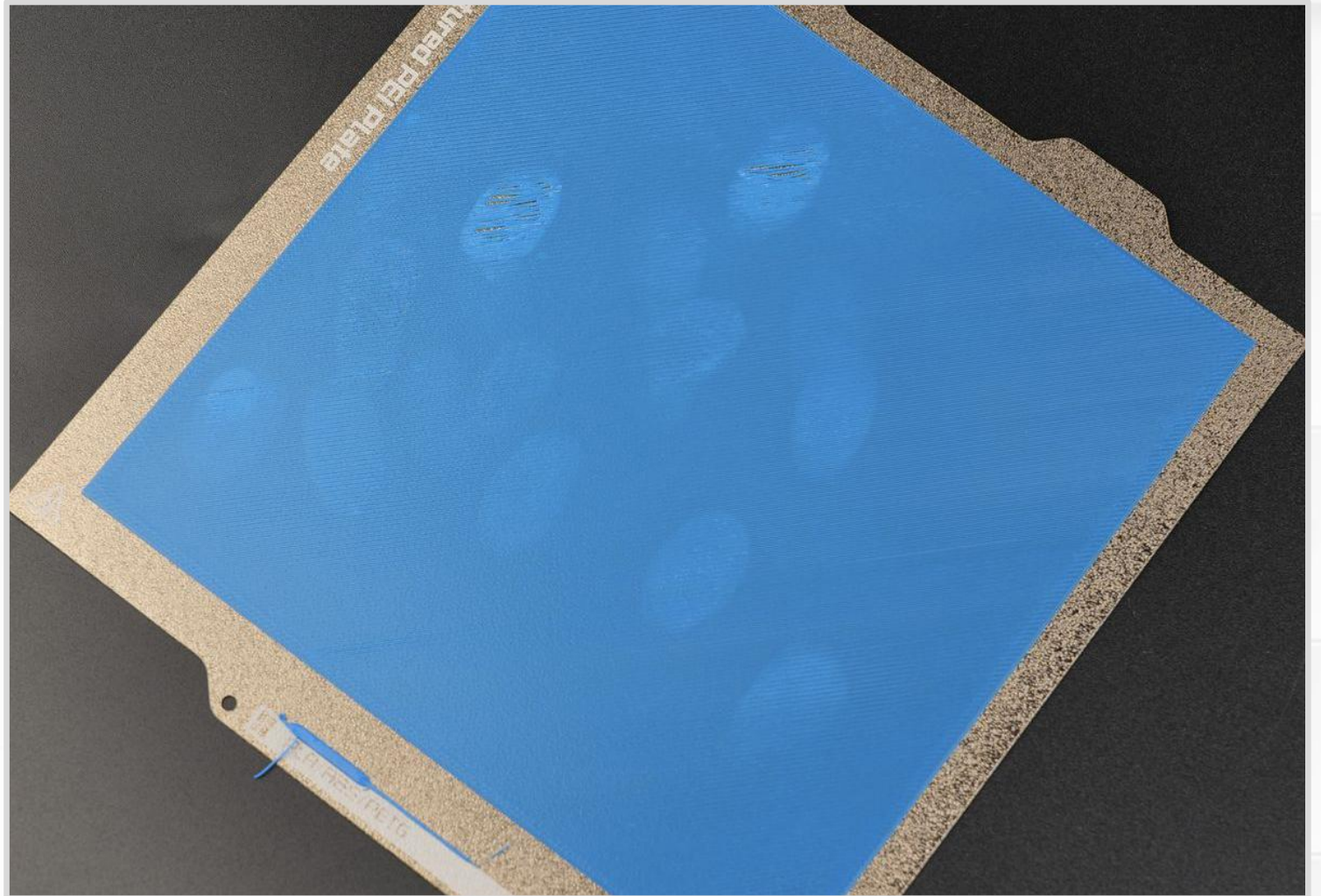
- What is wrong with this picture?
- **Fingerprints!** Plastic does not stick to oily fingerprints.



(Picture from [Bambu Labs Wiki](#))

# First Layer Mistakes

- Advice?
- Deep cleaning: dish soap, soft brush, and water at the sink.
- Everyday use: window cleaner with ammonia.



(Picture from [Bambu Labs Wiki](#))



# First Layer Mistakes

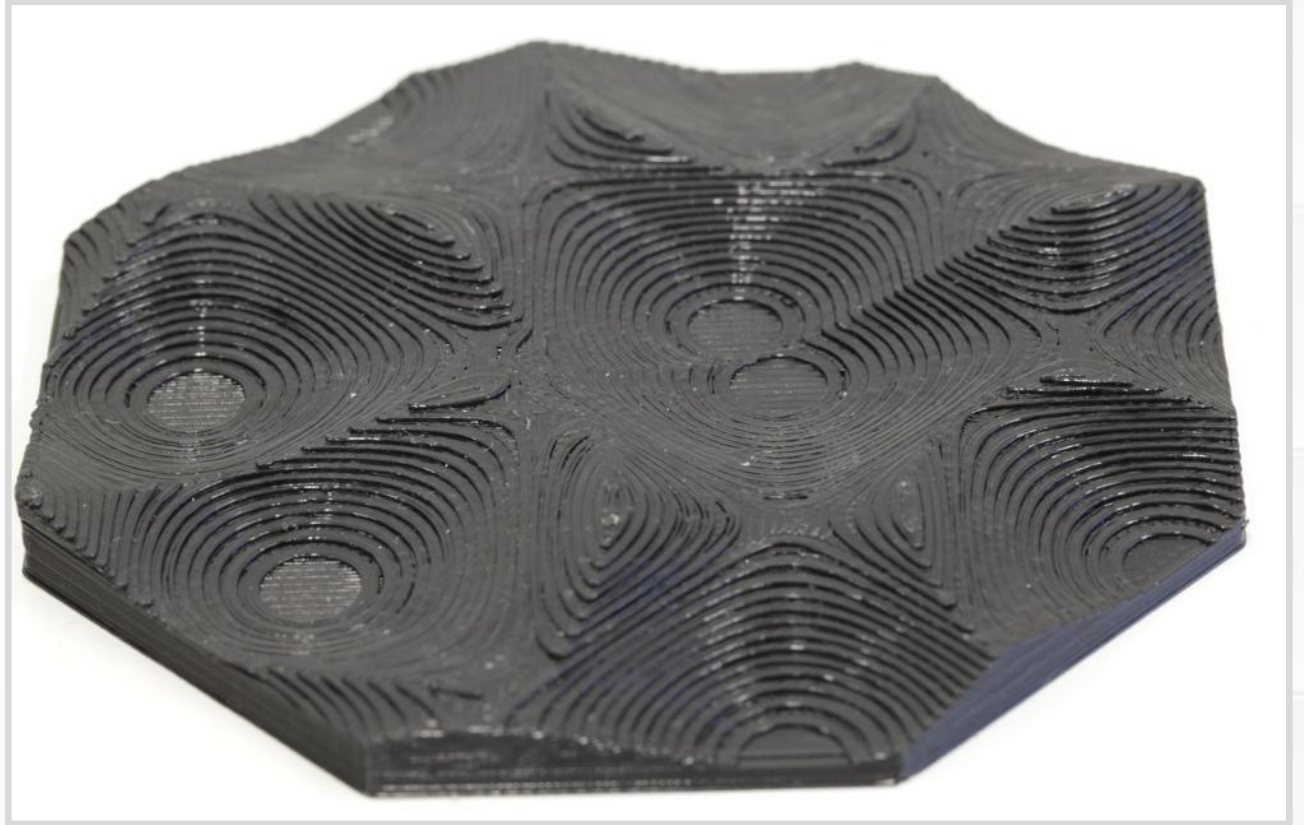
- What happens when a first layer does not stick?
- Print breaks free and makes “spaghetti”.
- When did things go wrong?
- Look at the partially completed, rectangular print.



(Screenshot from Everything STEM)

# Stair-Stepping

- Printing with layers → Vertical slopes turn into stair steps.
- Mostly cosmetic effect.
- More pronounced on shallow angles.
- Advice?
- Rotate objects to hide layer lines.
- Smaller layer height at the cost of increased print time.



(Picture from Hamburg University Research Project)

# First Layer Warping

- Why do corners and edges peel up?
- Regions shrink at different rates → Internal stresses → Stresses concentrate in corners.
- Shrinking is a percent → Large parts have the most problems.



(Picture from [Creality Cloud Blog](#))

# First Layer Warping

Consider this analogy.

- Imagine you want to remove a label or piece of tape.
- What do you do?
- Start peeling up the corner because that is the weak point.
- Same thing here.
- Physics found the weak point.



(Picture from [Creality Cloud Blog](#))



# First Layer Warping

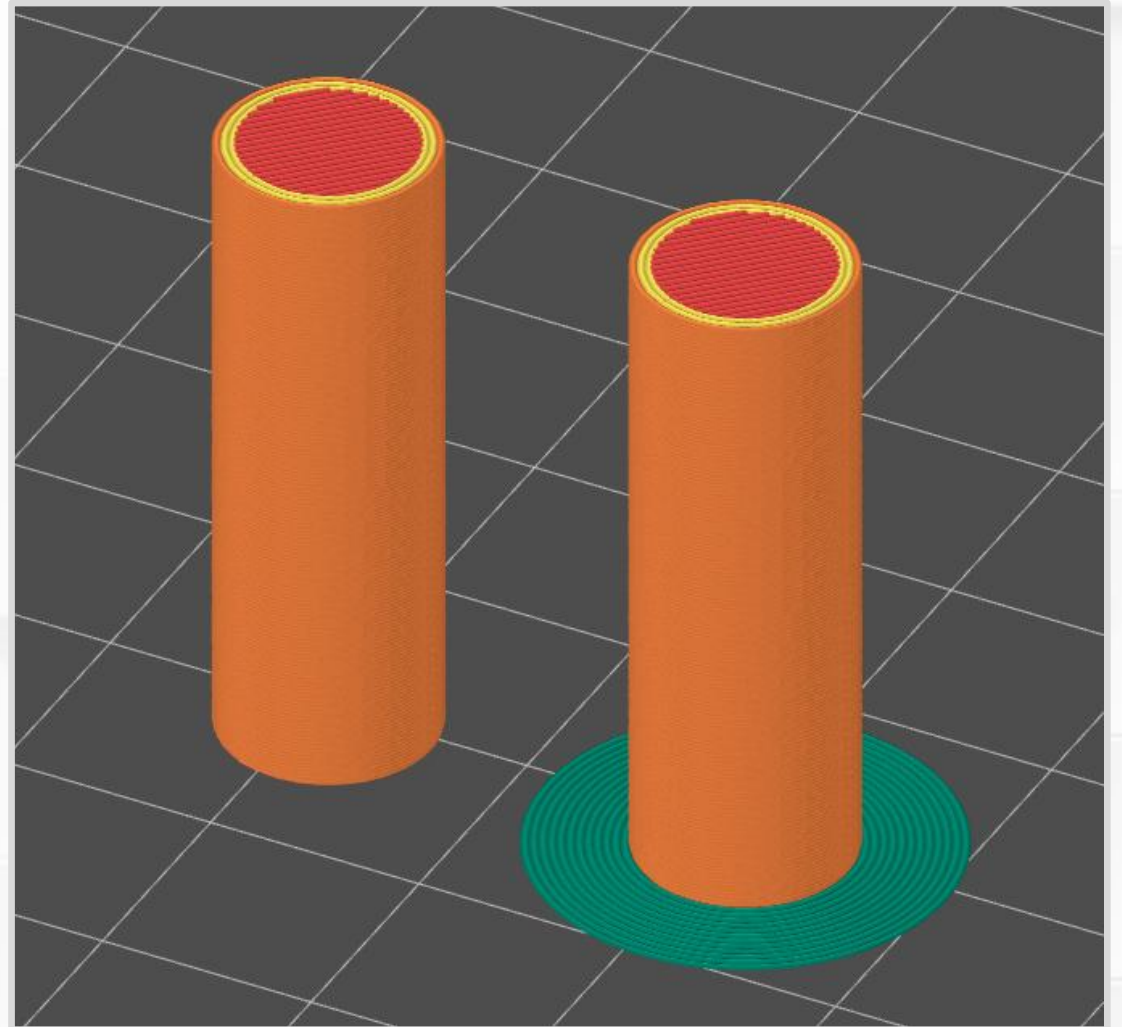
- Advice?
- Better control over temperatures.
  - Filament temperature.
  - Print chamber temperature.
  - Cooling fans.
- Evenly distributed cooling.
  - Avoid fans or drafts that cool one side.
- Round bottom corners to reduce stress concentration.



(Picture from [Crealitty Cloud Blog](#))

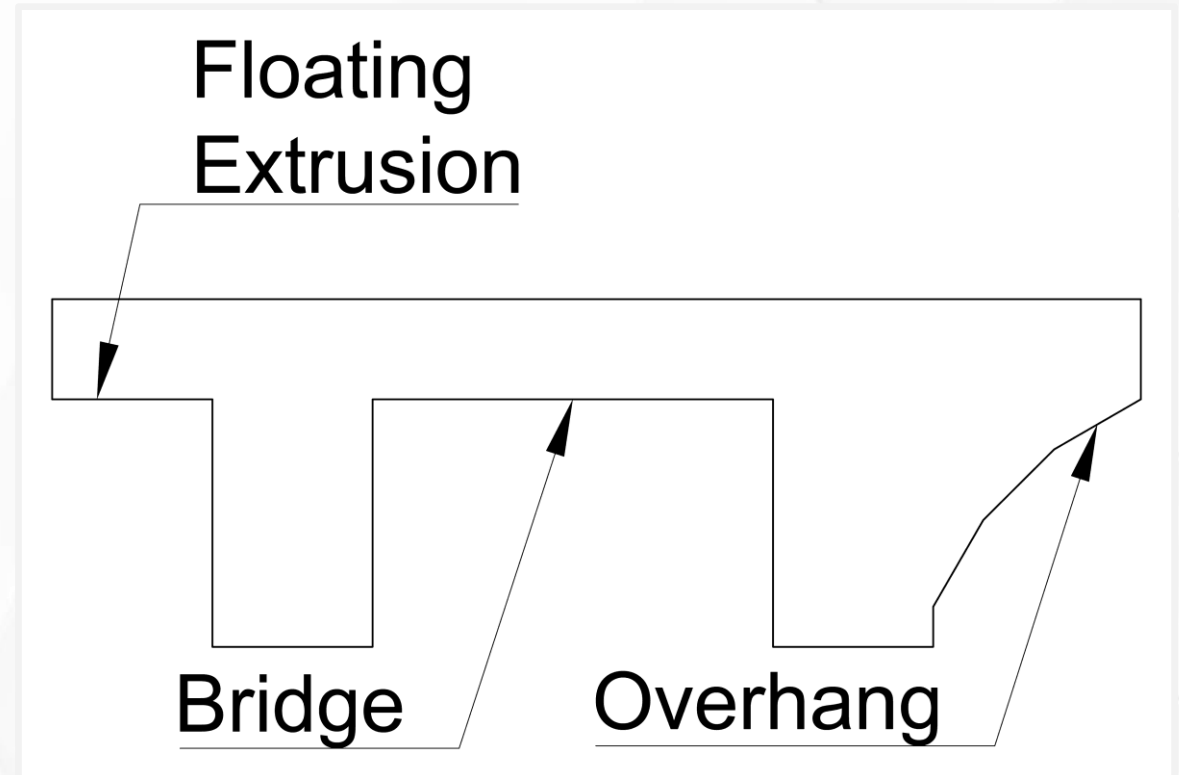
# Brim

- What if you could add extra surface area to fight warping?
- Brim, like the brim on a hat.
- Concentric extrusions joined to bottom perimeter. (Green lines.)
- Added with a slicer.
- One layer tall.
- Easily peeled away after printing.



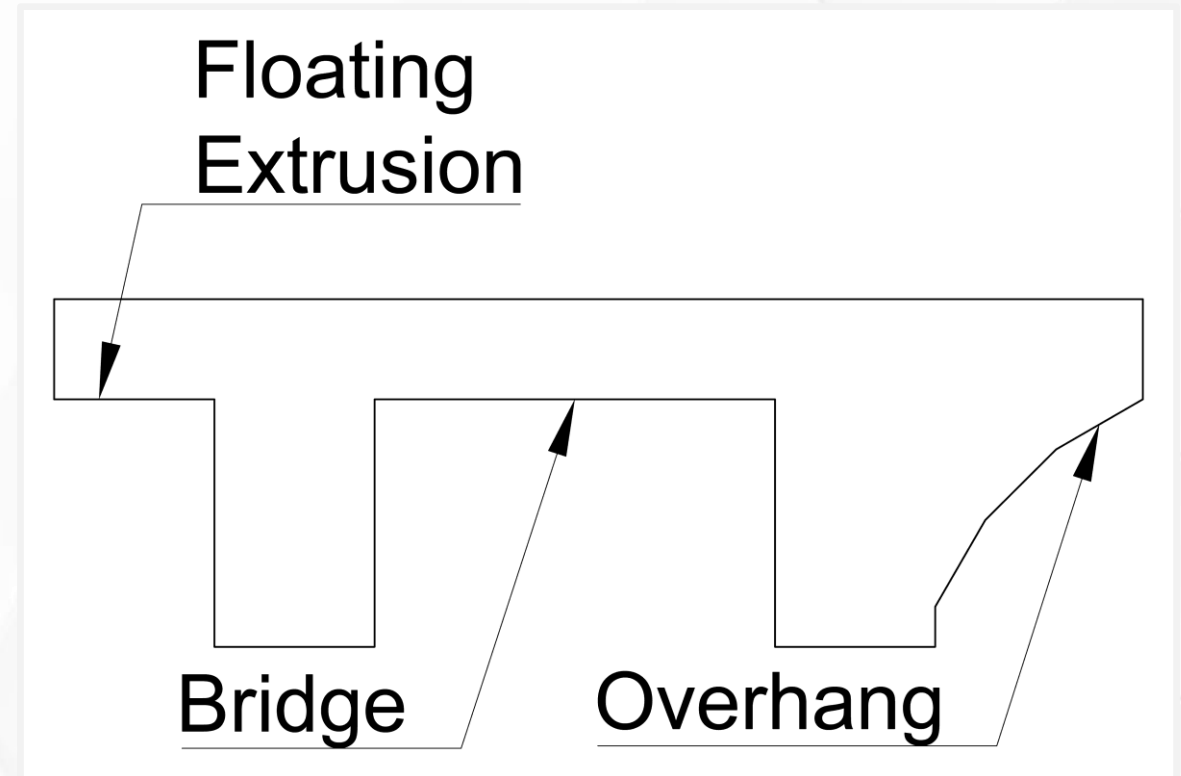
# Levitating Extrusions

- *Critical thinking:* What happens when you try to levitate extrusions?
- Extrusions work best when you press them onto a solid surface.
- Predict what this print will look like.
- What parts will be the most deformed?



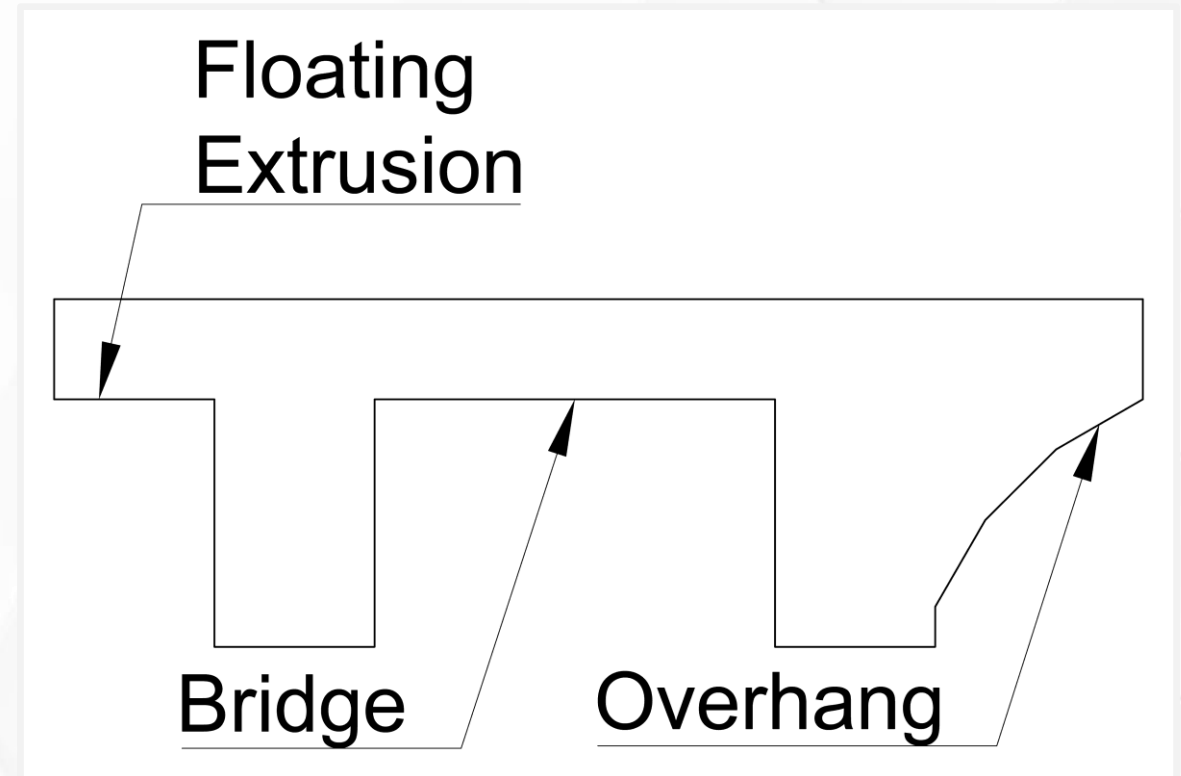
# Overhang

- Extrusions that lean over an edge.
- Only the outermost extrusion is affected.
- Partially supported along one side.
- Advice?
- Keep overhangs at  $45^\circ$  to be safe.



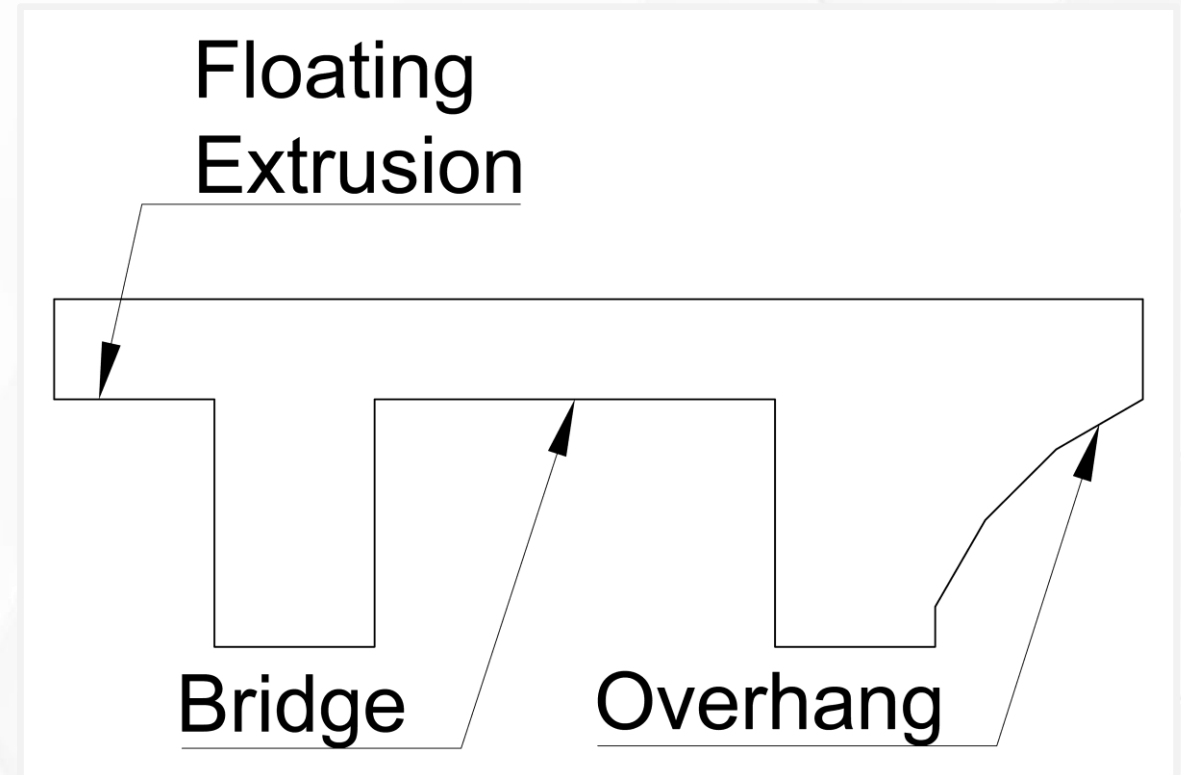
# Bridge

- Both ends of an extrusion are supported and anchored.
- Middle hovers over a gap.
- First few layers sag and droop down.
- Eventually recovers to flat layers.
- Advice?
- Keep bridges as short as possible.



# Floating Extrusion

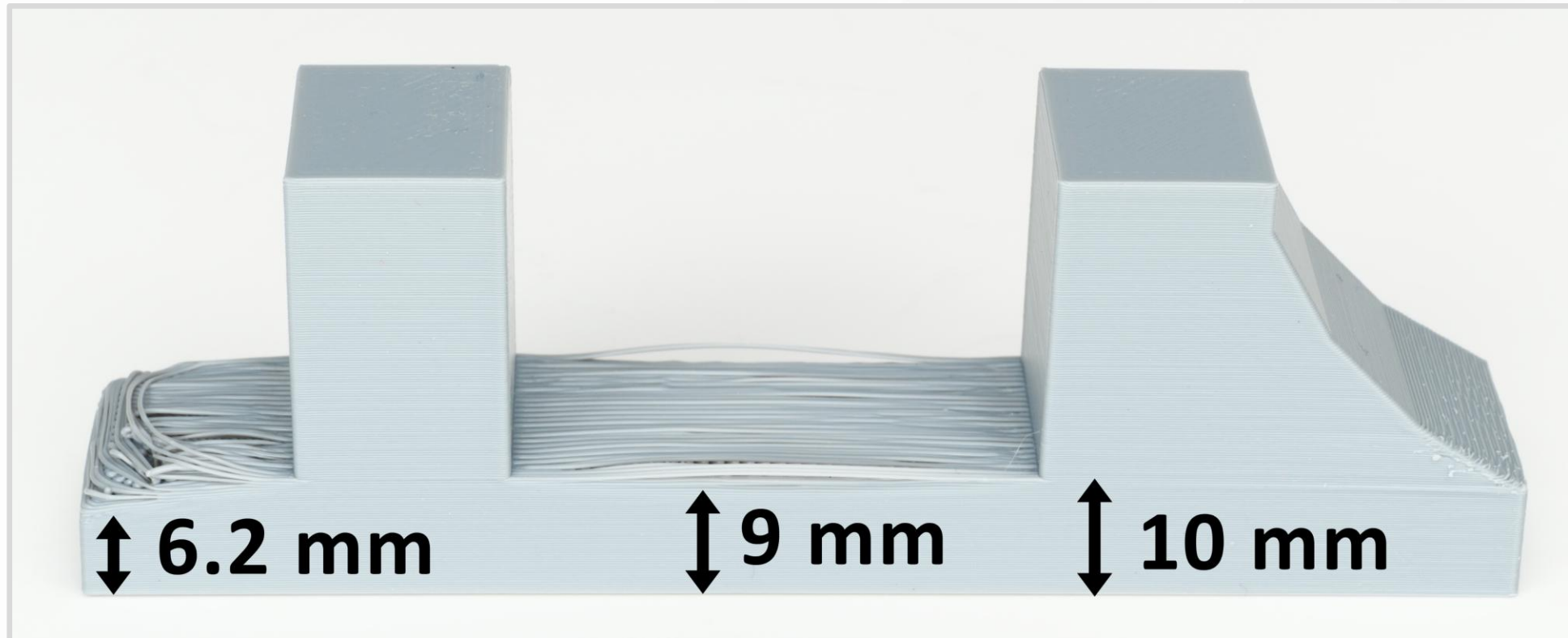
- Extrusion that hangs in the air.
- At least one end is not anchored to a solid surface.
- Advice?
- **Avoid. Very unreliable.**





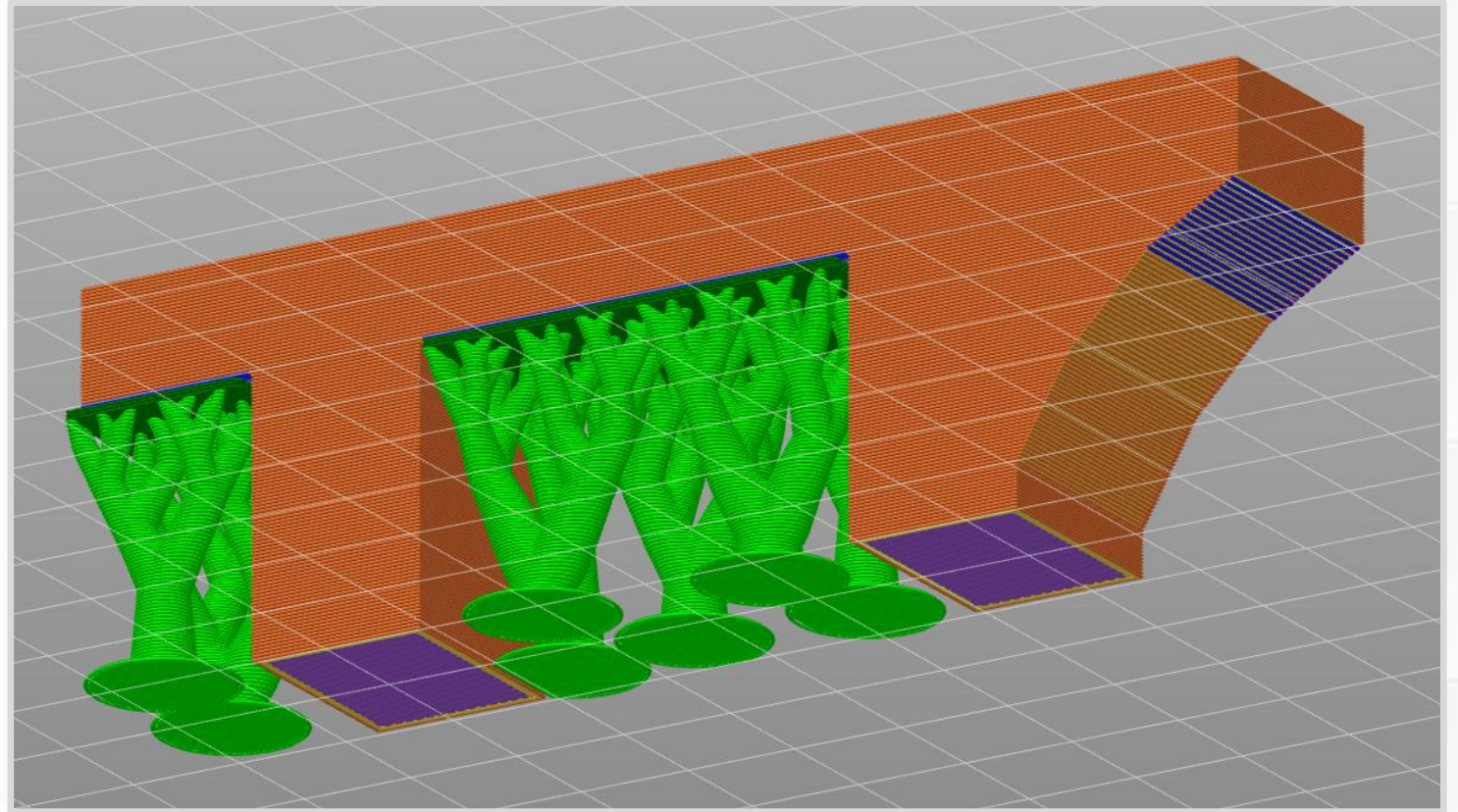
# Stress Test Results

- All overhang angles look good.
- Bridge recovers to flat layers in 1 mm (**5 layers**).
- Floating extrusions recover in 3.8 mm (**19 layers**).



# Supports

- Temporary scaffold that fixes:
  - Floating extrusions.
  - Long bridges.
  - Steep overhangs.
- Added by the slicer.  
(Green trees-like branches.)
- Extra complexity, plastic, and print time.

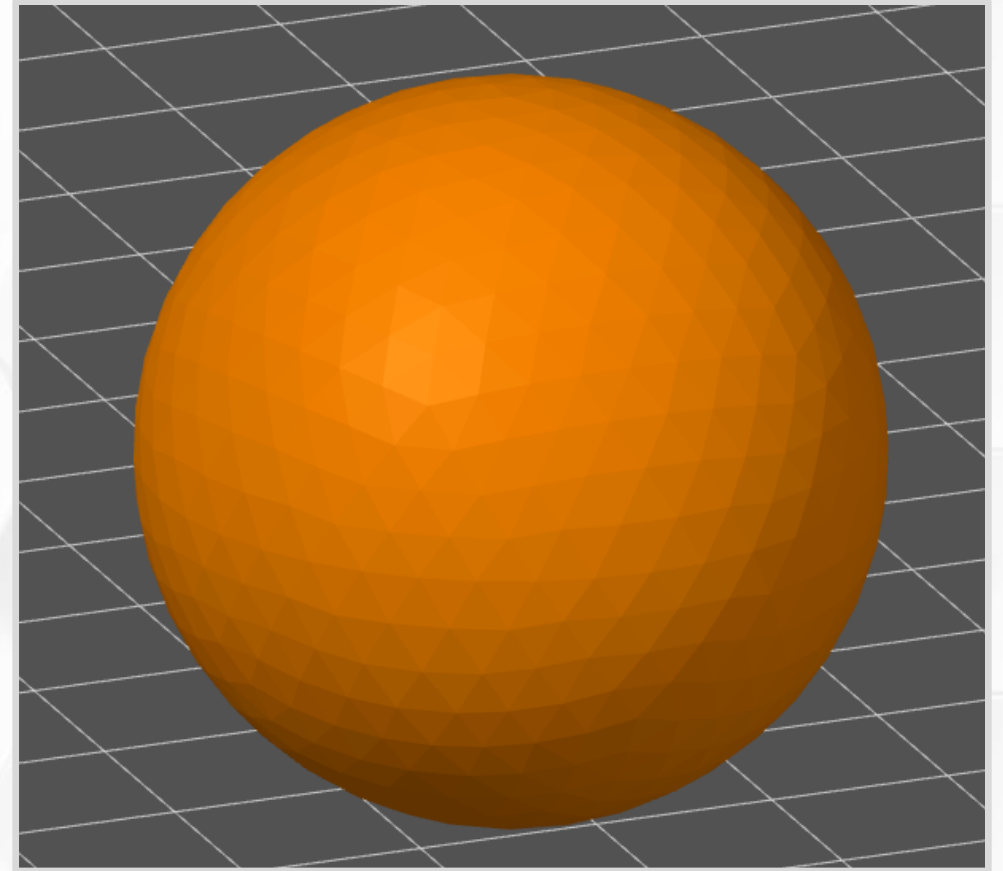


# Supports

- In special situations, you can include supports as part of a 3D model.
  - *Example:* 3D Printing Trick by Clough42, and a follow-up video.
- Tree/organic supports can reach into difficult spaces.
- Supports are a delicate balance. Why?
  - If they do not bond well enough, they hold nothing in place.
  - If they bond too well, they become part of the print.
- Supports can leave a rough surface finish.
- Sometimes you need to cut away pieces of support.

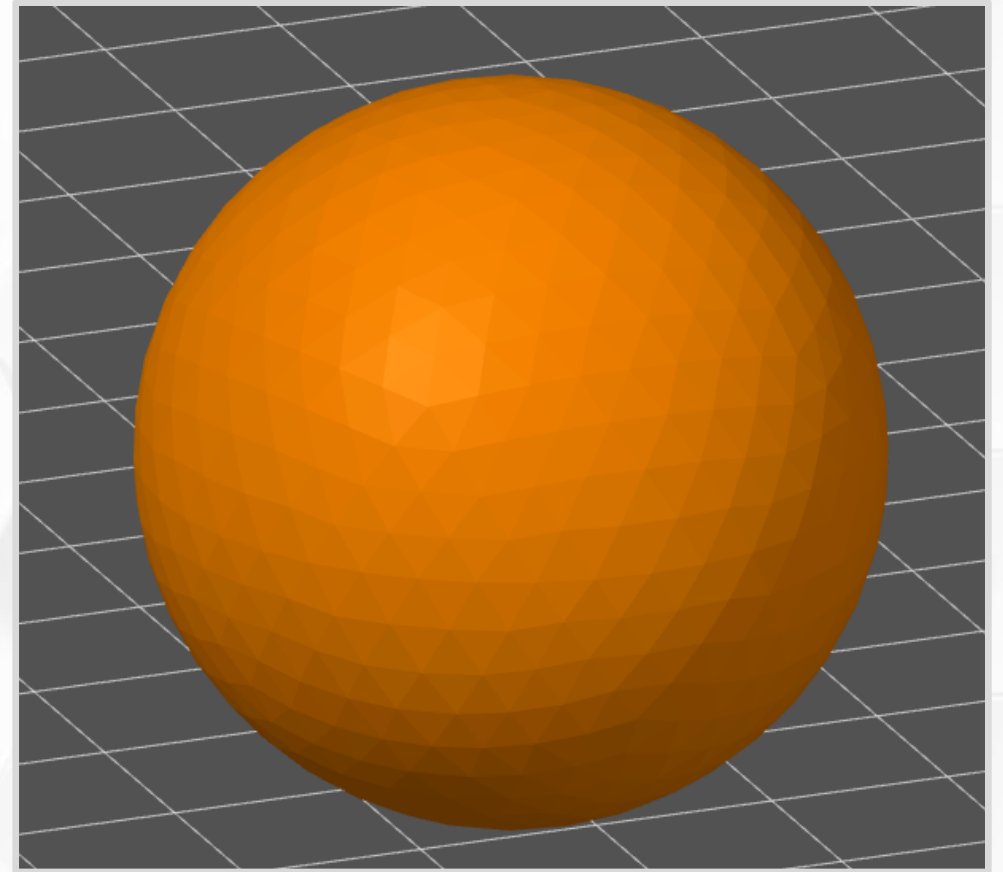
# Limitations Review

- **Printability** summarizes limitations for a specific object.
- Without extra considerations, a simple sphere has poor printability.
- Why? Make a list of limitations.



# Limitations Review

- In order of importance ...
- Very bad first layer contact. Significant spaghetti risk.
- Steep overhangs on the bottom.
- Second layer is steep enough to create floating extrusions.
- Uneven surface finish at the top and bottom.
- How can we fix this print?



# Limitations Review

- Brim is not helpful.
  - First layer is too small.
- Add supports to the bottom.
- Cannot fix stair-stepping by changing print orientation.
  - Use small layer heights to reduce visual artifacts.
  - Use adaptive layer heights to make the surface look more uniform.
- Print two halves, then put them together.
  - Add tabs to help you align them.

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Thermal Expansion



# Dimensional Accuracy

- Be careful modeling parts with exact dimensions.
- Why?
  - 3D prints are imperfect.
  - When assembled together, the fit is probably too tight.
- Where do these errors come from?
- And how do we fix it?



# Dimensional Accuracy

- Sources of error include but are not limited to:
  - Parts that shrink or warp.
  - Inconsistent extrusions.
  - Racked gantry (skewed axes).
  - Shipping damage.
  - Incorrect belt tension.
  - Backlash, slop, or loose parts.
  - Moving parts that are worn down.
  - Sloppy slicer settings.
  - Thermal expansion (*discussed in Part 3*).
  - Vibrations (*discussed in Part 3*).

# Modeling Calibration

- *(Printed size) = (Size) × (Scale factor) + (Constant offset).*
- Simplified model.
  - Assume fully functional motion system.
  - Apply to X and Y dimensions.
- Scale factor. Error changes with size.
  - Part shrinkage.
- Constant offset. Error is always the same.
  - Extrusion inconsistencies. Over and under-extruding.
  - Surface roughness. Additives like carbon fiber, wet filament, over-extruding.



([xkcd.com](http://xkcd.com))

# Fixing Scale Factor

- **Adjust part scaling in the slicer to fix scale factor.**
  - Scale XY size, or
  - Dedicated shrinkage compensation setting.
- Applies to
  - Large parts over 6 inches, 150 mm.
  - Filaments that shrink, like ABS.
- In my experience, safe to ignore for PLA, PETG, and small projects.

# Fixing Constant Offset

- **Add clearance (air gap) between parts that touch to fix constant offset.**
- Very important for projects with assembled pieces!
- Best to add clearance while designing 3D objects.
  - Greatest flexibility. Independently adjust each surface.
- Or try adding negative XY offset in slicer.
  - Equally offsets all vertical surfaces.

# Adding XY Clearance

- How much XY clearance? It depends ...
- Start with 0.25 mm. Offset each side by half, 0.125 mm.
- *Example:* Making a hole for a bolt.
  - Add 0.2 - 0.3 mm of clearance for a close fit.
- *Example:* Making a lid that easily slides into place.
  - Add 0.4 - 0.8 mm of clearance for a loose fit.

# Adding Z Clearance

- How much Z clearance? It depends ...
- Start with one layer or 0.1 mm.
- Increase clearance in certain scenarios.
  - Sagging extrusions.
  - Rough bottom surface from supports.
  - High and low spots on print bed.
  - Tilted print bed.



# Summary

If your project has assembled parts then ...

- You sometimes need scale factor.
- You always need constant offset.



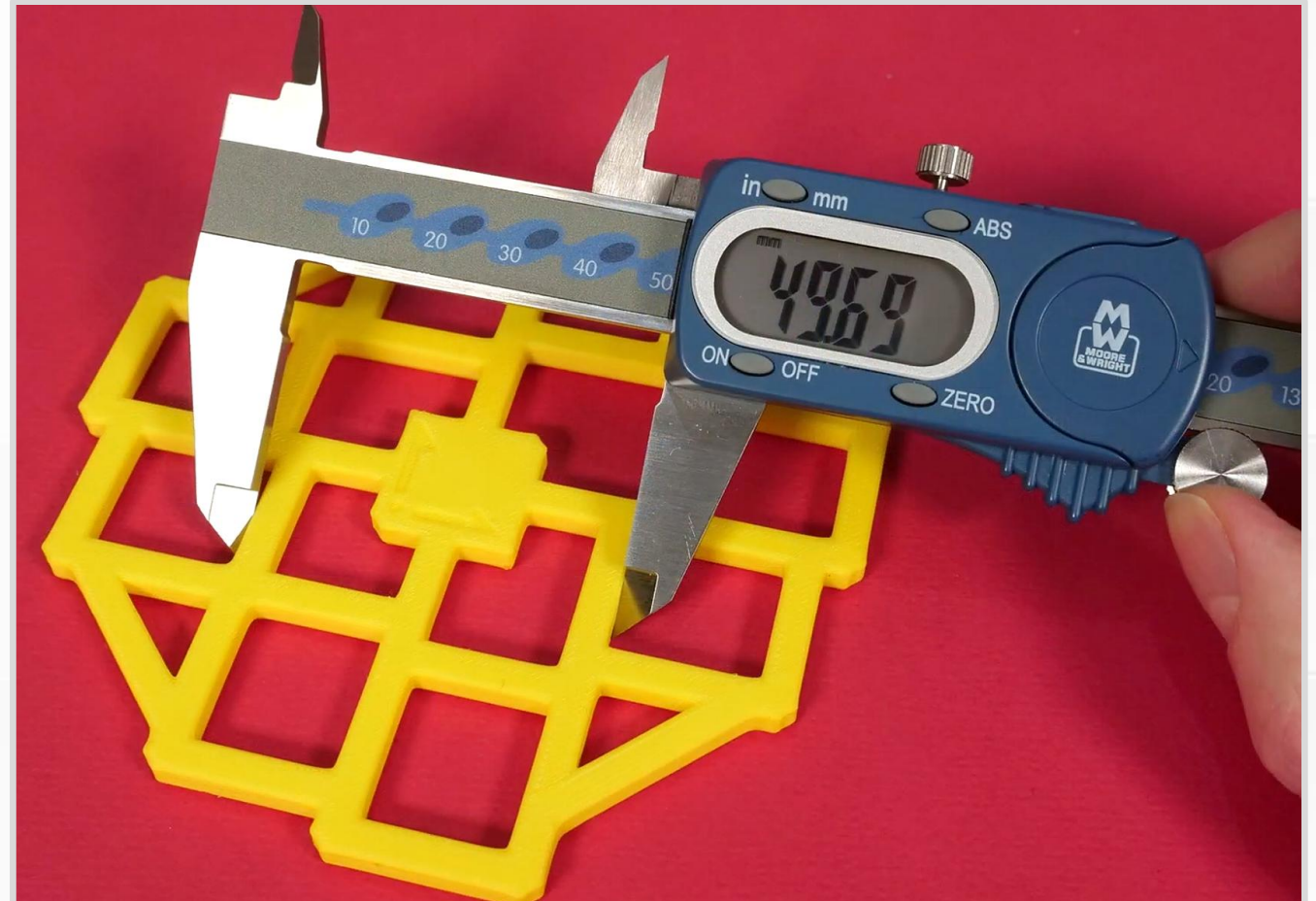
# Calibration Prints

- Calibration cubes.
  - 20 mm cube.
  - Labels for X and Y-axis.
  - Tells you constant offset for outside dimensions.
- Clearance test by 3DMakerNoob.
  - Print-in-place clearance/tolerance.
  - Alters the gap between pieces.
  - Feel which pieces move freely.



# Calibration Prints

- Califlower by Vector3D.
  - Reasonably priced. Includes walkthrough and calculator.
  - Inner dimensions.
  - Outer dimensions.
  - Shrinkage.
  - Skew, when rectangles turn into parallelograms.
- Calibration video by CNC Kitchen.



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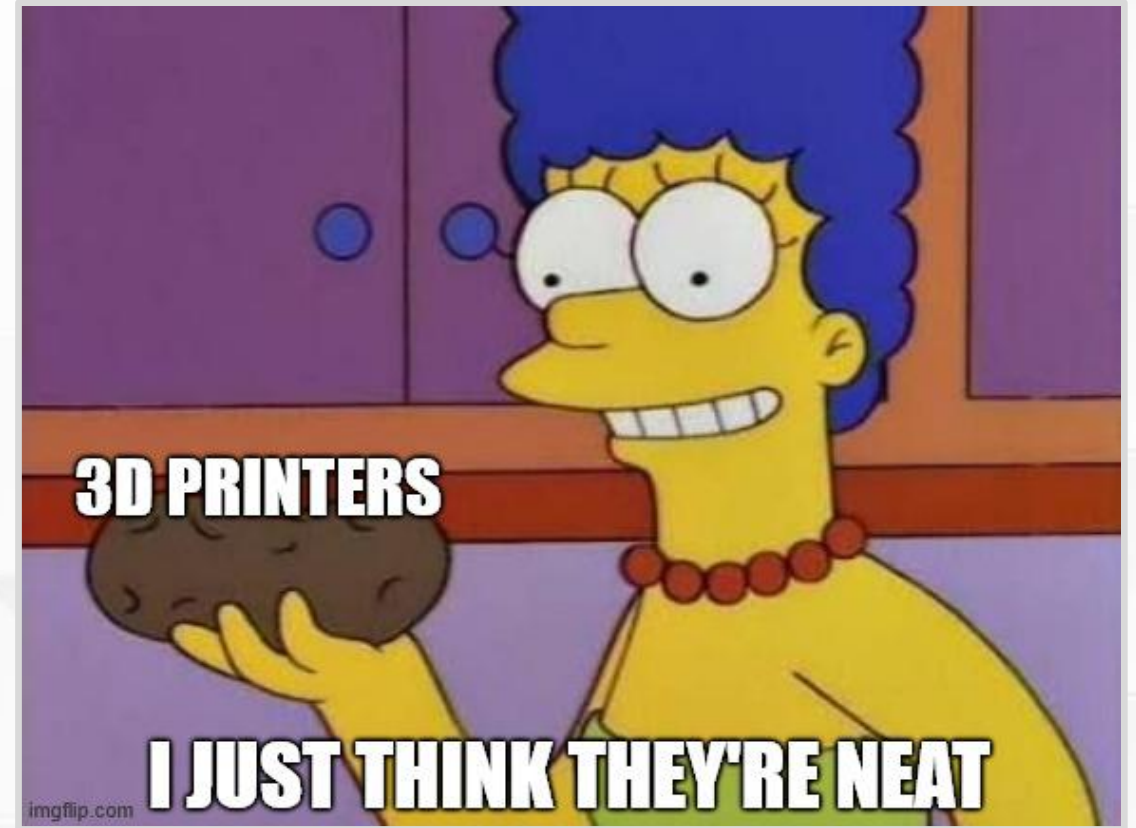
Linear Motion

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# An Amazing Tool

- Playground for **creative design** and **creative problem solving**.
- Massive utility for cheap.
- Make replacement parts to fix and repair.
- Bring your ideas to life.
- Fast turnaround time.
- Low barrier to entry.
- Practical for hobbyists, students, and professionals.

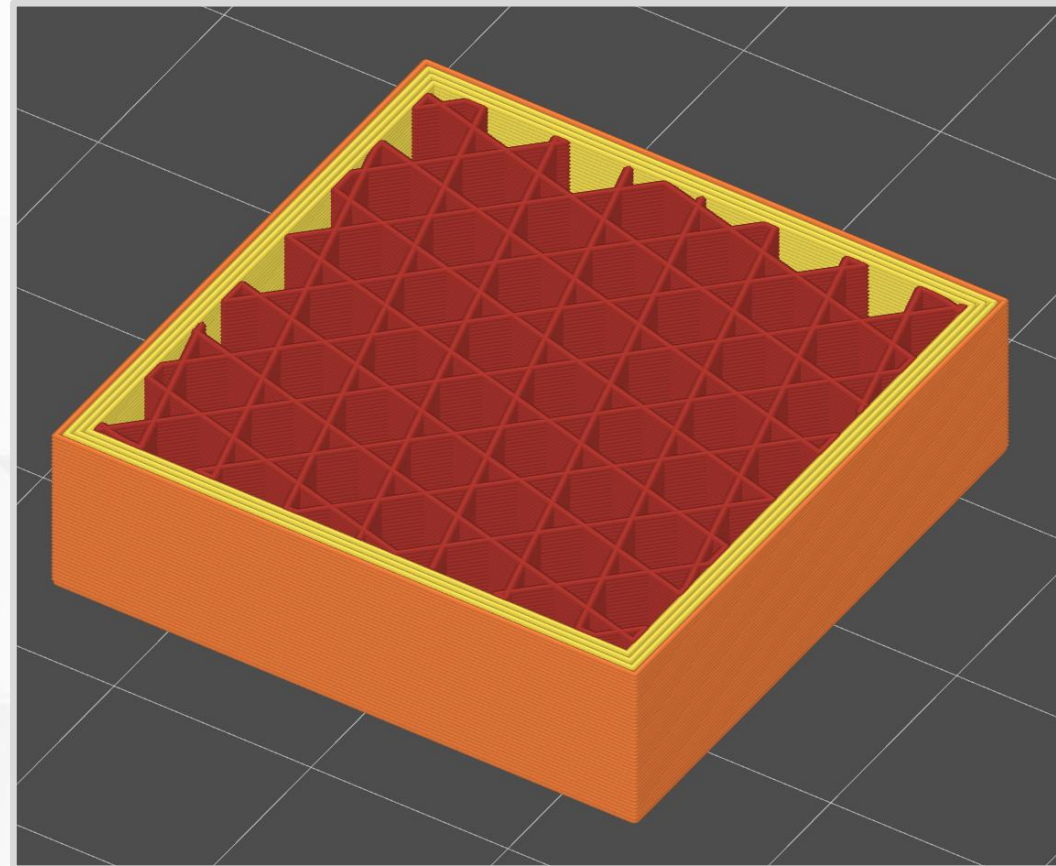


# An Amazing Community

- RepRap festivals, maker fairs, local clubs and maker spaces, ...
- People share cool projects and ideas. Examples include:
  - Full spectrum printing*
  - BumpMesh.com*
  - Voron*
- YouTube makers with educational content. Examples include:
  - CNC Kitchen*
  - Maker's Muse*
  - JanTec Engineering*
  - 3D Printing Nerd*
  - Teaching Tech (retired)*

# FFF Advantages

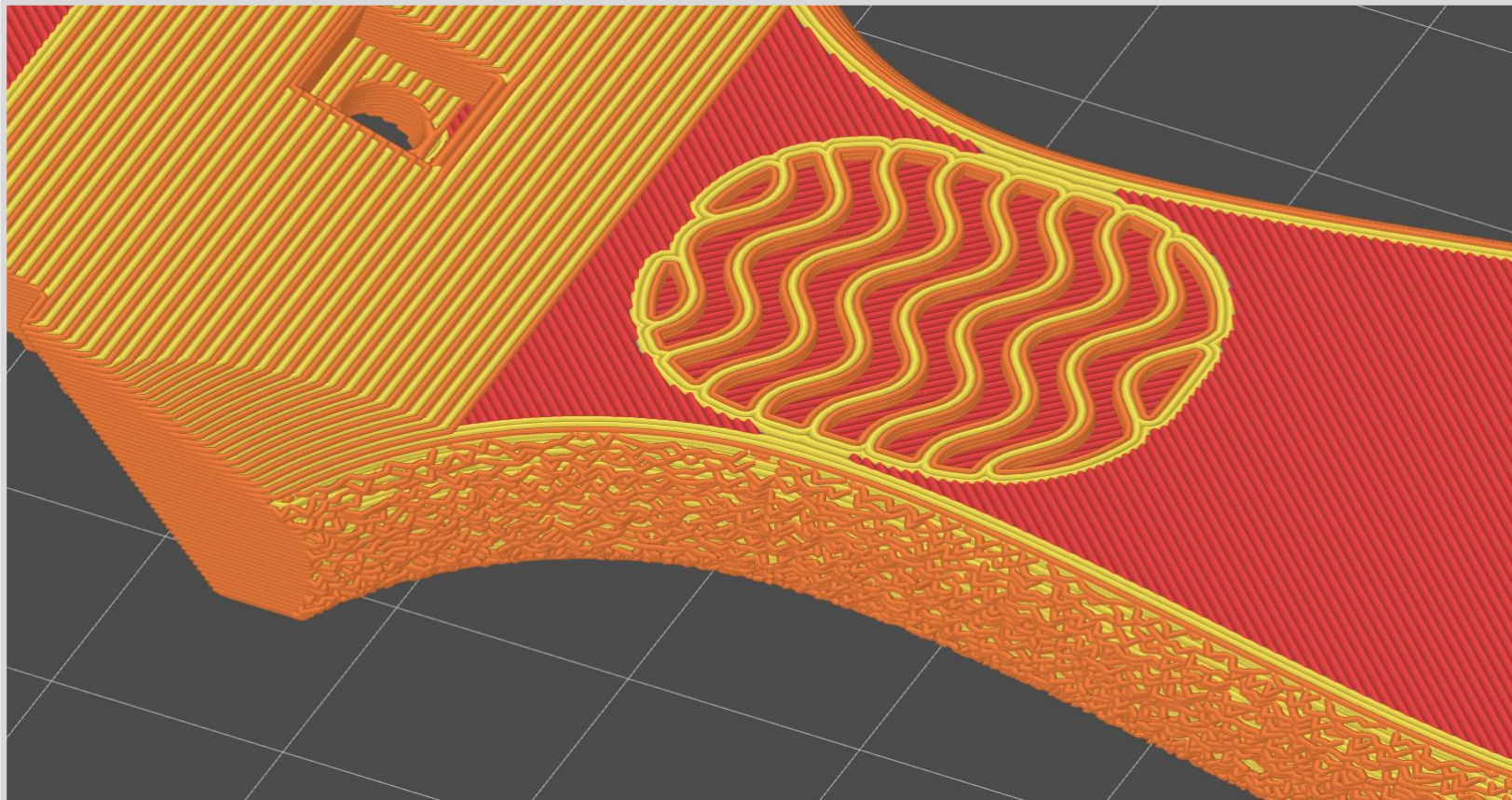
- Make lots of shapes with tiny “building blocks.”
  - Typical extrusions are  
**0.2 mm tall** (1-2 pieces of paper)  
**0.4 mm wide**.
- Not restricted to uniform, solid objects.
  - Easily create internal geometry.
  - Great at making hollow objects.





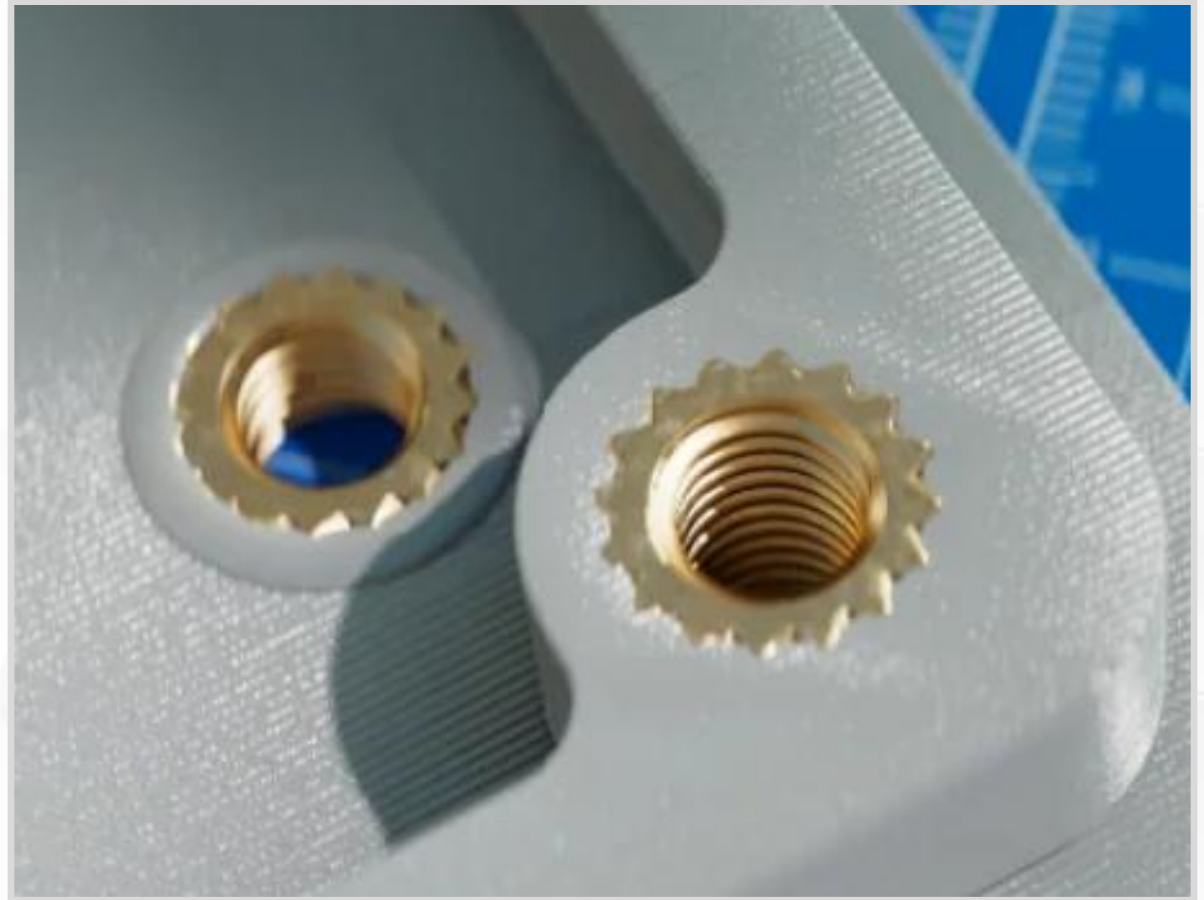
# Custom Slicing

- Change slicer settings for specific regions or surfaces. Too many to list.
- *Example:* Add texture to handles with fuzzy skin (bumpy extrusions).



# Heat Set Inserts

- Threaded anchor for screws. Why?
- Plastic is soft → Screws easily wear out plastic holes.
- Install with a hot soldering iron.
- Locks in place when it cools.
- Very durable.



(Pictures from [CNC Kitchen Store](#))

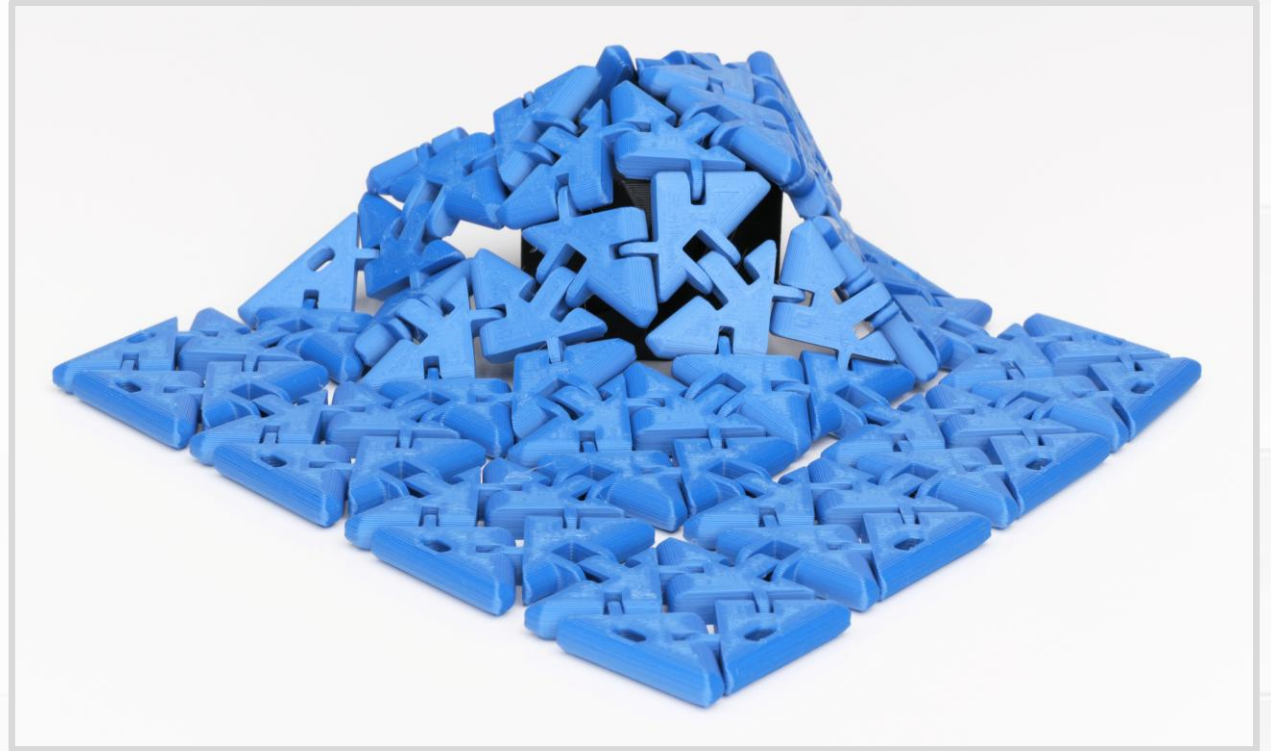
# Captive Parts

- Insert an object into a pocket or void in the middle of a print.
  - Nuts and fasteners.
  - Magnets.
  - Weights.
  - Reinforcing bar or plate.
- Locked inside when the print finishes.
- Object **must** sit below the current layer, or it creates a collision with the toolhead.
- A little inconvenient.
  - Need to be at the printer at the right layer.



# Print-in-Place (PIP)

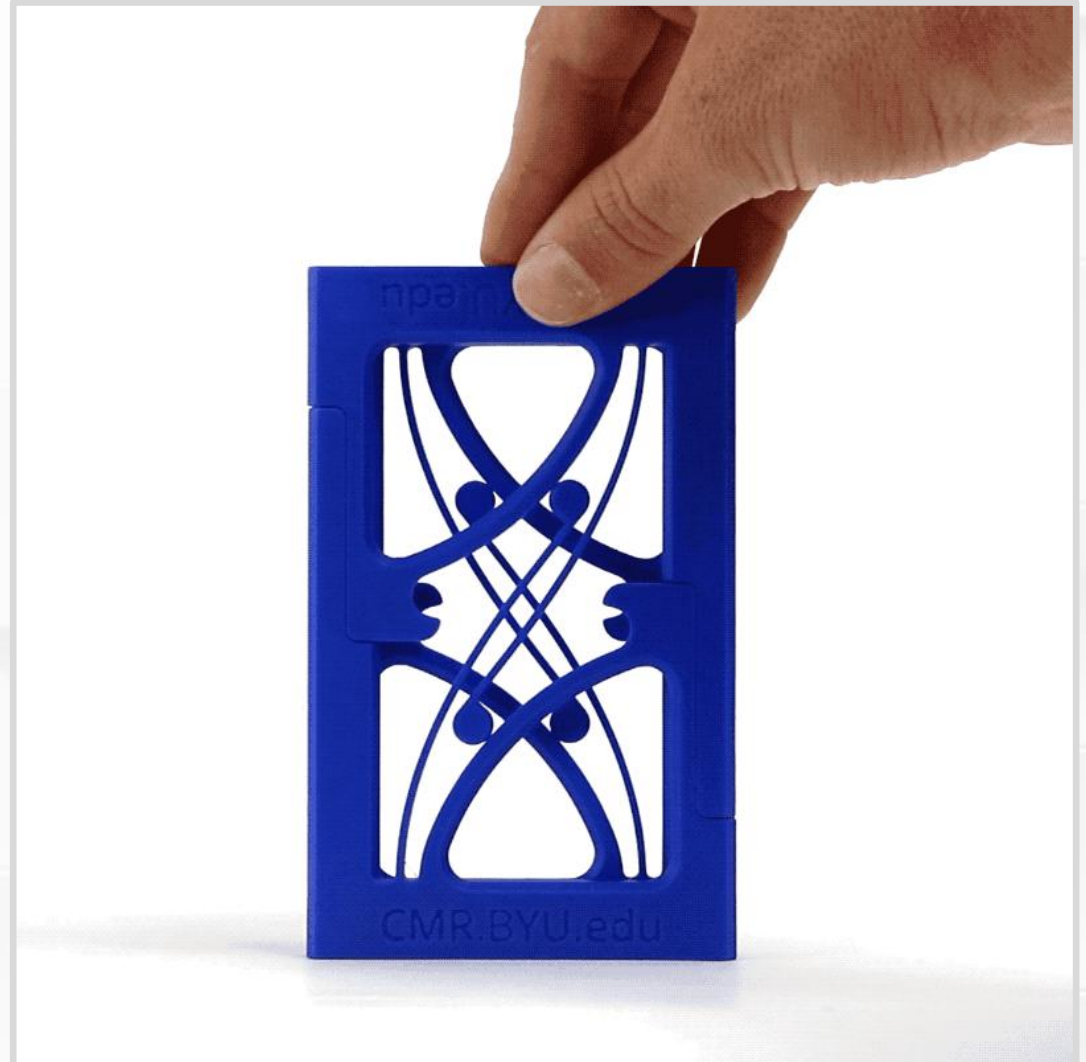
- Single print with interlocking parts.
- Assembled when the print finishes.
- *Example:* Hinges, chain links, chainmail.
- Flexify collection by [printables.com/@Flying Gyroscope](https://printables.com/@FlyingGyroscope)



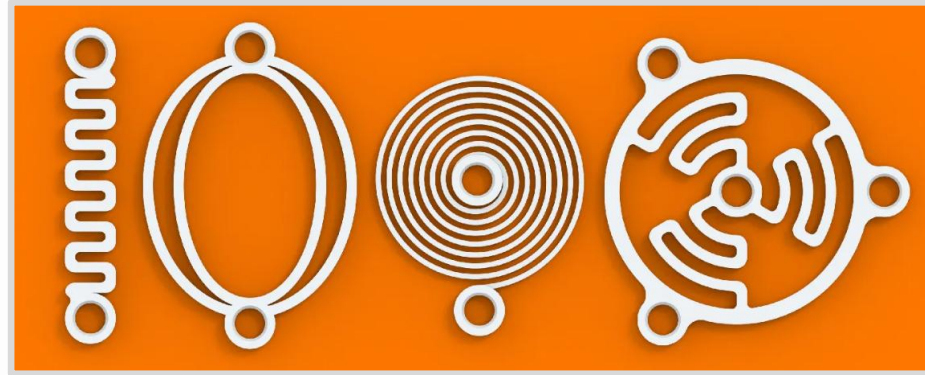
# Functional Flexing

- Springs and **compliant mechanisms**.
- Features and shapes that function by bending and flexing.
- Elastic, springy materials.
  - No permanent deformation.
- Remember part strength and layer lines.

(Animation, Constant Force  
Mechanism by BYU CMR)



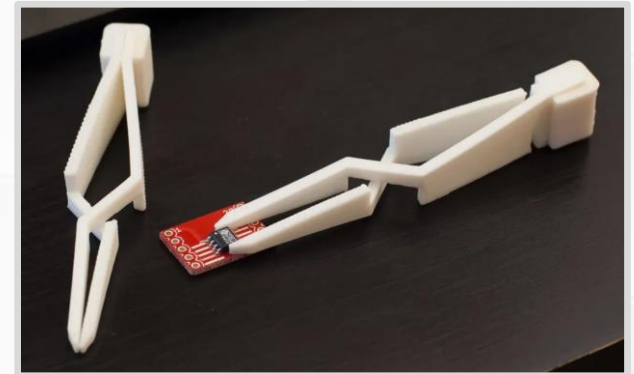
Springs by VC Design



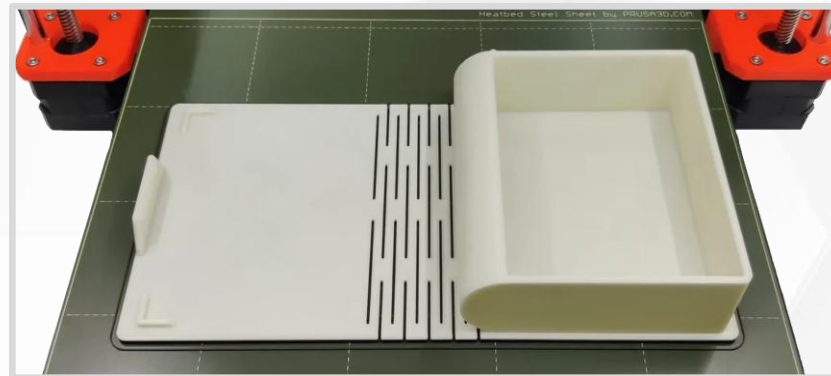
Clip by AE



Cross tweezer by joo



Living hinge by DrJones





[github.com/FlyingGyroscope/IntroTo3DPrinting/](https://github.com/FlyingGyroscope/IntroTo3DPrinting/)